Quenched Convergence of GMC-Sampled Tilings in Every Dimension $d \geq 3$

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Abstract

For every integer $d \geq 3$ and sufficiently small $\gamma > 0$, we consider the conductance walk on the Euclidean Voronoi tessellation sampled from γ-Gaussian multiplicative chaos of an α-quantum cone, where $\gamma Q - \gamma^2/2 = d$ and $\alpha Q - \alpha^2/2 = 2$. A deterministic time $c_{d,\gamma}\varepsilon^{\alpha/\gamma}$ per step gives convergence to α-Liouville Brownian motion. The conditional laws given the field and the tiling converge in probability. We integrate the normalizer, stationarity, ergodicity, geometric, and clock arguments in one manuscript. The odd-dimensional extension uses the previously constructed quantum cones and a descending-record Palm argument. Localizing the Stieltjes record measure proves exterior-tail triviality even in dimension three, without differentiating its sphere-average process or restarting at a negative first-passage time.

Status. This is an unrefereed AI-assisted research manuscript. Its Gaussian cone construction and standard GMC inputs are stated explicitly below. Lean formalization is not supplied here.

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1 Model and main theorem

Fix an integer $d \geq 3$ and put

$$0 < \gamma < \sqrt{2d} - \sqrt{2d-1}, \qquad Q = \frac{d}{\gamma} + \frac{\gamma}{2}, \qquad \alpha = Q - \sqrt{Q^2 - 4}, \qquad \vartheta = \frac{\alpha}{\gamma}, \quad \kappa = Q - \alpha. \tag{1.1}$$

Thus $0 < \alpha < \gamma$ and

$$\gamma Q - \frac{\gamma^2}{2} = d, \qquad \alpha Q - \frac{\alpha^2}{2} = 2, \qquad 2 = d\vartheta + \frac{\gamma^2}{2}\vartheta(1-\vartheta). \tag{1.2}$$

The field h is the α-quantum cone in the unit-sphere first-passage embedding: its radial part at $|x| = e^{-t}$ is $S^*_t + \alpha t$, and its independent spherical part has zero average on every centered sphere. The process S^* is the first-passage limit recalled in Section 2. In even dimensions this is the cone of Bertacco–Gwynne [1]; in odd dimensions it is the cone constructed in [2].

For $\beta \in \{\gamma, \alpha\}$ use the local measurable spherical-average normalization

$$\mu_{h,\beta}(dx) = \lim_{r \downarrow 0} r^{\beta^2/2} e^{\beta h_r(x)}\, dx, \qquad M_h = \mu_{h,\gamma}, \quad \nu_h = \mu_{h,\alpha}.$$

The first measure samples nuclei; the second is the speed measure. Conditionally on h, let $\mathcal{P}_\varepsilon \sim \mathrm{PPP}(\varepsilon^{-1} M_h)$. A nucleus v has closed Euclidean Voronoi cell $\overline{P}^\varepsilon_v$. Independent continuous marks assign distance ties to give a measurable partition P^ε_v. Boundaries have zero ν_h-mass. Write $v \sim w$ for a common facet of positive $(d-1)$-dimensional measure and define

$$a_\varepsilon(v,w) = \frac{\mathcal{H}^{d-1}(\overline{P}^\varepsilon_v \cap \overline{P}^\varepsilon_w)}{|v-w|}, \qquad \pi_\varepsilon(v) = \sum_{w \sim v} a_\varepsilon(v,w).$$

The walk starts at the nucleus o_ε of the cell containing 0 and jumps from v to w with probability $a_\varepsilon(v,w)/\pi_\varepsilon(v)$. For unit intensity omit ε and set

$$Z = \mathbb{E}\left[\frac{\pi(o)}{\nu_h(P_o)}\right]. \tag{1.3}$$

Theorem 1.1. *Under (1.1), $0 < Z < \infty$. Set $c_{d,\gamma} = 2/Z$ and linearly interpolate the walk, assigning the interval $[j c_{d,\gamma}\varepsilon^\vartheta, (j+1)c_{d,\gamma}\varepsilon^\vartheta]$ to its jth step. Denote this path by X^ε. Let X^h be α-Liouville Brownian motion in h from 0, with underlying Brownian generator $\frac{1}{2}\Delta$. Then*

$$d_{\Pr}(\mathrm{Law}(X^\varepsilon \mid h, \mathcal{P}_\varepsilon), \mathrm{Law}(X^h \mid h)) \longrightarrow 0, \tag{1.4}$$

$$d_{\Pr}(\mathrm{Law}(X^\varepsilon \mid h), \mathrm{Law}(X^h \mid h)) \longrightarrow 0. \tag{1.5}$$

The first convergence is in joint probability over the field and point process; the second is in probability over the field. The laws are on $C([0,\infty), \mathbb{R}^d)$ with the local uniform topology. The step paths have the same conditional limits in the local Skorokhod J_1 topology.

This theorem is for the small-parameter range (1.1), for an α-cone sampled with γ-GMC. It is not a full-subcritical-range assertion or an almost-sure quenched limit along all intensities. The normalizing constant depends on both GMC normalizations in the displayed definition, and the factor 2 fixes the generator convention.

For $z \in \mathbb{R}^d$, $r > 0$, write $\mathcal{S}_{z,r}(h, \mathcal{P}) = (h(z + r\,\cdot) + Q \log r, r^{-1}(\mathcal{P} - z))$. GMC covariance gives

$$\mu_{h(z+r\cdot)+Q\log r,\beta}(A) = r^{\beta Q - \beta^2/2 - d} \mu_{h,\beta}(z + rA). \tag{1.6}$$

Thus M pulls back without a factor, while ν and all conductances are multiplied by r^{2-d}. Adding a constant b multiplies M by $e^{\gamma b}$ and ν by $e^{\alpha b}$.

Dependencies and organization

We use the odd-cone construction and its full-half-line Gaussian comparison estimate from [2], and their even-dimensional counterparts from [1]. The Palm identification, exterior-tail argument, and their $d = 3$ transfer are proved here. Standard GMC moment and insertion results are from [7, 6, 8]. We prove the needed finite-graph elliptic estimate, following [3, Theorem 5.4] with an explicit local mesh correction. We then derive the normalized joint exit kernels and their path convergence; convergence modulo time in [4] would not suffice. The strict-clock and bridge theory is from [1, 5], with the cone-root and odd-dimensional adaptations given below. No constant-speed walk convergence or decorated-cone ergodicity is taken as an assumption.

2 Radial level stationarity and the exterior tail

Fix an integer $d \geq 3$ and $\kappa > 0$. Let $S_0 = 0$ be the sphere-average process of the normalized whole-space log-correlated Gaussian field, with covariance $-\log|x-y|$ modulo constants and unit-sphere average zero. The moving-average construction is given by [1, Theorem A] in even dimensions and [2, Theorem A] in odd dimensions:

$$S_t = \int_0^\infty q_d(u)(B_{t-u} - B_{-u})\,du, \qquad q_d(u) = (d-2)e^{-2u}(1-e^{-2u})^{(d-4)/2}.$$

Set $K_d(u) = (1-e^{-2u})^{(d-2)/2}$ for $u \geq 0$. We use one conditional-future framework in every dimension. There is a state space E_d and a continuous state process $A_t \in E_d$ such that, at every deterministic $r \geq 0$, the conditional law of $(S_{r+s})_{s\geq 0}$ given its driving-Brownian past is the future kernel initialized at (S_r, A_r). Write $P_u^{x,a}$ for that kernel restricted to the entire future after elapsed time u, with coordinates indexed from u. The negative Brownian history is included in the filtration at time zero. The state space and kernel are specified in the following proof; every subsequent argument uses their common properties only.

Lemma 2.1 (Full-future state forgetting)**.** *For every $d \geq 3$ there is a bound $\Delta_d(u, L) \in [0, 1]$, nonincreasing in u and nondecreasing in L, such that*

$$d_{\mathrm{TV}}(P_u^{x,a}, P_u^{y,b}) \leq \Delta_d(u, L) \quad \text{if } |x| + \|a\|_{E_d} + |y| + \|b\|_{E_d} \leq L, \qquad \lim_{u\to\infty} \Delta_d(u, L) = 0. \tag{2.1}$$

The completed future and remote-past tails of the unconditioned process S are trivial in every dimension $d \geq 3$.

Proof. If $d = 2m + 1$ is odd, take $E_d = E_{\lambda,m}$, $0 < \lambda < 2$, from [2, Definition 5.2], and $A_t = M_t$ from its Lemma 5.3. Its Proposition 5.4 identifies the conditional future as

$$x + \int_0^s K_d(s-v)\,dW_v + M(s) - M(0), \qquad s \geq 0,$$

where W is independent of the past. Choose a smooth ψ supported in $[0, 1)$ with $\psi(0) = 1$. For $u \geq 3$, adding height x is exactly replacing the memory M by $M - x\psi$. Proposition 5.9 of [2] therefore gives

$$d_{\mathrm{TV}}(P_u^{x,M}, P_u^{y,N}) \leq \min\{1, C(u^{-1/2} + e^{-\lambda u})(|x-y| + \|M-N\|_{E_d})\}.$$

This proves (2.1) in odd dimensions and verifies continuity of the required state paths.

If $d \geq 4$ is even, put $p = (d-2)/2$ and take $E_d = \mathbb{R}^p$ with $A_t = (S'_t, \dots, S^{(p)}_t)$. By [1, Theorem A], (S_t, A_t) is the integrated Gaussian Markov process of [1, Lemma 3.14], whose derivative drift matrix has distinct negative eigenvalues. That lemma gives total-variation forgetting of the state at time u, uniformly over bounded initial states. Applying its common Markov future kernel does not increase total variation and gives exactly the full-future comparison (2.1). These state paths are continuous. Taking the supremum over the indicated bounded initial states gives a bound monotone in L; full-future restriction makes it nonincreasing in u.

To prove the tail assertion, fix $r \geq 0$ and compare the conditional future initialized at (S_r, A_r) with an independent reference initialized at its state at r. Average (2.1) over the reference state. Almost-sure finiteness of the two states and dominated convergence show that the conditional law of $(S_t)_{t \geq n}$ approaches its unconditional law in total variation as $n \to \infty$, almost surely. For a completed future-tail event F, choose for each integer n a representative measurable from $(S_t)_{t \geq n}$. The comparison implies $\mathbb{P}(F \mid \mathcal{F}_r) = \mathbb{P}(F)$. Martingale convergence as integer $r \to \infty$ makes its indicator constant. Gaussian stationary increments give $(-S_{-t})_{t \in \mathbb{R}} \overset{\text{law}}{=} S$, so time reversal proves the remote-past assertion. □

Put

$$X_t = S_t - \kappa t, \qquad I_t = \inf_{s \leq t} X_s, \qquad \nu((s,t]) = I_s - I_t, \qquad (\theta_t S)(u) = S_{t+u} - S_t.$$

The Brownian moving-average formula gives the decomposition $Z_t = -\int_{-\infty}^{t} (1 - K_d(t-s))\, dB_s$ and $S_t = B_t + Z_t - Z_0$, where Z is a stationary continuous Gaussian process. In particular $S_t/|t| \to 0$ almost surely in both time directions. Thus I is finite and continuous, and ν is a diffuse locally finite Stieltjes measure. It is shift covariant: $\nu_{\theta_r S}(A) = \nu_S(r + A)$.

Lemma 2.2 (Record-Palm identification and level stationarity)**.** *The cone radial limit $S^\star$ from [1, Theorem C] in even dimensions and [2, Theorem B] in odd dimensions has law*

$$\mathbb{E}^\star F(S^\star) = \frac{1}{\kappa} \mathbb{E} \int_{(0,1]} F(\theta_t S)\, \nu(dt) \tag{2.2}$$

for every bounded measurable function on the continuous path space. Moreover $\mathbb{E}\nu((0,1]) = \kappa$, $S^\star_t/|t| \to 0$ in both time directions, and $S^\star_t - \kappa t > 0$ for every $t < 0$, almost surely.

For $a \in \mathbb{R}$, put

$$\tau_a = \inf\{t \in \mathbb{R} : S^\star_t - \kappa t = -a\}.$$

Then τ_a is finite and

$$\theta_{\tau_a} S^\star \overset{\text{law}}{=} S^\star. \tag{2.3}$$

Proof. The drawdown $D_t = X_t - I_t$ is stationary by shift covariance. Time reversal gives

$$D_0 \overset{\text{law}}{=} \sup_{t \geq 0} (S_t - \kappa t).$$

This supremum is integrable: the decomposition of S bounds it by

$$\sup_{t \geq 0} (B_t - \kappa t/2) + \sup_{t \geq 0} (Z_t - \kappa t/2) + |Z_0|.$$

The Brownian term has an exponential tail; the stationary Gaussian term is integrable by the uniform sub-Gaussian bound for $\sup_{n \leq t \leq n+1} |Z_t|$ and summation over integers $n \geq 0$. Consequently

$$\mathbb{E}\nu((0,T]) = \mathbb{E}(I_0 - I_T) = \mathbb{E}(X_0 - X_T) = \kappa T.$$

The right side of (2.2) therefore defines a probability law, temporarily denoted P^{rec}.

For bounded measurable F, the sequence

$$Y_k = \int_{(k,k+1]} F(\theta_t S)\,\nu(dt), \qquad k \in \mathbb{Z},$$

is stationary and ergodic: it is a shift-equivariant factor of two-sided white noise, whose time-one shift is ergodic. It is integrable since $|Y_k| \le \|F\|_\infty \nu((k,k+1])$. Birkhoff's theorem gives

$$\frac{1}{T}\int_{(0,T]} F(\theta_t S)\,\nu(dt) \longrightarrow \kappa \mathbb{E}^{\mathrm{rec}} F \quad \text{almost surely.} \tag{2.4}$$

This holds for arbitrary real $T \to \infty$: the error from the last unit interval is bounded by $\|F\|_\infty \nu((n,n+1])/n$, which tends to zero almost surely by integrability of the common marginal and Borel–Cantelli.

Let $J = -I$ and $T_b = \inf\{t : J_t \ge b\}$. The continuous nondecreasing function J tends to $-\infty$ and $+\infty$ at the two ends of the line. For $b > J_0$, its inverse T_b equals the global first-passage time $\sigma_b = \inf\{t : S_t - \kappa t = -b\}$. Inverse substitution for continuous Stieltjes measures gives

$$\int_{J_0}^{b} F(\theta_{\sigma_y} S)\,dy = \int_{(0,T_b]} F(\theta_t S)\,\nu(dt), \qquad \nu((0,T_b]) = b - J_0.$$

Applying (2.4) first with $F = 1$ shows $T_b/b \to 1/\kappa$. Applying it with the original F then yields

$$\frac{1}{b}\int_0^b F(\theta_{\sigma_y} S)\,dy \longrightarrow \mathbb{E}^{\mathrm{rec}} F \quad \text{almost surely.} \tag{2.5}$$

The bounded interval $[0, J_0]$ contributes a quantity tending to zero. For bounded continuous local F, the stated radial local-limit theorem gives $\mathbb{E}F(\theta_{\sigma_y} S) \to \mathbb{E}F(S^\star)$. Taking expectations in (2.5), using bounded convergence, and comparing these Cesàro limits proves $P^{\mathrm{rec}} = \mathcal{L}(S^\star)$, since such local functions determine laws on the continuous path space.

Two-sided sublinearity is preserved by every finite recentering, so it passes through (2.2). The measure ν is carried by record times $X_t = I_t$. For ν-almost every such t, one also has $I_s > I_t$ for all $s < t$. Indeed, exceptions belong to interiors or endpoints of nontrivial flat intervals of I; there are countably many maximal such intervals, their interiors have zero ν-measure, and their endpoints have zero mass since ν is diffuse. This proves the asserted strict past positivity under P^{rec}.

For the final assertion, let R_a be the measurable map that recenters a path at its first global hit of the drifted level $-a$. The pathwise identity

$$R_a(\theta_{\sigma_y} S) = \theta_{\sigma_{y+a}} S$$

holds for all real y, a. Apply (2.5) to F and $F \circ R_a$. Their level integrals differ, after translating the level variable, by at most $2|a|\|F\|_\infty/b$. Their limits agree. This proves (2.3) for every bounded measurable F, without a continuity assertion for hitting-time maps or a conditional law at a negative cone passage. □

Proposition 2.3 (Unconditional first-passage scaling)**.** *In every dimension $d \ge 3$, let h be the β-cone with $\kappa = Q - \beta > 0$. For $a \in \mathbb{R}$, let τ_a be as in Lemma 2.2 and put $r_a = e^{-\tau_a}$. Then*

$$h(r_a\,\cdot) + Q \log r_a + a \overset{\mathrm{law}}{=} h. \tag{2.6}$$

Proof. The radial identity is (2.3), since $S^\star_{\tau_a} - \kappa\tau_a = -a$. The independent spherical projection is invariant under deterministic dilations. Since r_a is radial-measurable, its randomly dilated spherical projection retains that law and its independence from the radial part. This proves the full field identity. □

Theorem 2.4 (Asymptotic agreement of radial exterior laws)**.** *Let P be the law of S and $P^\star$ the law in (2.2). With $\mathcal{A}_T = \sigma(\omega(t) : t \le -T)$ on the canonical path space,*

$$\lim_{T\to\infty} d_{\mathrm{TV}}(P^\star|_{\mathcal{A}_T}, P|_{\mathcal{A}_T}) = 0. \tag{2.7}$$

Consequently $\bigcap_{n\ge1} \overline{\mathcal{A}_n}^{P^\star}$ is trivial, where the bar denotes completion under $P^\star$.

Proof. For $R \ge 1$ and $0 \le t \le 1$, define

$$I^R_t = \min_{-R\le s\le t} X_s, \qquad \nu^R((s,t]) = I^R_s - I^R_t \quad (0 \le s < t \le 1).$$

On $(0,1]$ the measure ν^R dominates ν. Indeed, each running minimum is the minimum of its initial floor and the same running minimum over $[0,t]$; every drop of that minimum increases when the initial floor increases. As $R \uparrow \infty$, the initial floor decreases to I_0, so

$$\nu^R((0,1]) \downarrow \nu((0,1]), \qquad \nu^R((0,1]) \le \kappa + \sup_{0\le t\le1} |S_t|.$$

The last bound is integrable. Consequently, with $\kappa_R = \mathbb{E}\nu^R((0,1])$,

$$\mathbb{E}\|\nu^R - \nu\|_{\mathrm{TV}} = \kappa_R - \kappa \longrightarrow 0. \tag{2.8}$$

Define a probability law Q_R by

$$Q_R(F) = \frac{1}{\kappa_R}\mathbb{E}\int_{(0,1]} F(\theta_t S)\,\nu^R(dt).$$

Equation (2.8) implies $d_{\mathrm{TV}}(Q_R, P^\star) \to 0$ on the entire path space.

Fix R for the moment. By Gaussian time reversal, we may realize $V_v = -S_{-v}$ as another copy of the Brownian moving-average construction, with driving-Brownian filtration $\mathcal{G}_v$ and state A^V_v. The measure ν^R, the path $S|_{[-R,1]}$, and every V_{-t} for $t \in [0,1]$ are $\mathcal{G}_R$-measurable. For $v \ge T$,

$$-(\theta_t S)(-v) = V_{v-t} - V_{-t}. \tag{2.9}$$

Conditional on $\mathcal{G}_R$, the path on the right is, in the absolute observation variable v, initialized at time $R+t$ with height $x_t = V_R - V_{-t}$ and remaining state A^V_R. Its subsequent Brownian noise is independent of $\mathcal{G}_R$. This description holds simultaneously for $t \in [0,1]$, as follows directly from the state-driven pathwise construction in Lemma 2.1.

Let $(V', A^{V'})$ be an independent reference copy. Its unconditioned future, observed at the same absolute times v, is the mixture of the future laws initialized at time $R+t$ with states $(V'_{R+t}, A^{V'}_{R+t})$. For $T \ge R+4$, apply Lemma 2.1 with $u = T-R-t \ge T-R-1$. It gives a $\mathcal{G}_R$-measurable random bound $g_{T,R} \le 1$ such that

$$\sup_{0\le t\le1} d_{\mathrm{TV}}\left(\mathcal{L}((V_{v-t} - V_{-t})_{v\ge T} \mid \mathcal{G}_R), \mathcal{L}((V_v)_{v\ge T})\right) \le g_{T,R}, \tag{2.10}$$
$$g_{T,R} \le \mathbb{E}'\big[\Delta_d(T-R-1, H_R + H'_R)\big] \longrightarrow 0 \quad \text{almost surely},$$

where

$$H_R = |V_R| + \sup_{0\le t\le 1} |V_{-t}| + \|A_R^V\|_{E_d}, \qquad H'_R = \sup_{R\le s\le R+1} (|V'_s| + \|A_s^{V'}\|_{E_d}).$$

Both quantities are finite almost surely by path and state continuity. Equation (2.1) and dominated convergence prove the stated limit. Thus (2.10) controls both the random horizontal shift and the height subtraction in (2.9).

The random measure ν^R is $\mathcal{G}_R$-measurable. We can therefore integrate the conditional comparison against it, using the jointly measurable conditional kernel supplied by the pathwise construction. This is valid even if ν^R is singular with respect to Lebesgue measure. Taking the supremum over bounded test functions of the remote path yields

$$d_{\mathrm{TV}}(Q_R|_{\mathcal{A}_T}, P|_{\mathcal{A}_T}) \le \frac{1}{\kappa_R}\mathbb{E}[\nu^R((0,1])g_{T,R}] \longrightarrow 0 \qquad (T\to\infty). \tag{2.11}$$

Dominated convergence uses the integrable mass bound above. The triangle inequality, first letting $T\to\infty$ in (2.11) and then $R\to\infty$ in (2.8), proves (2.7).

Finally, let A belong to every $P^\star$-completed $\mathcal{A}_n$. Choose $A_n \in \mathcal{A}_n$ with $P^\star(A \triangle A_n) = 0$. For $n, m \ge N$,

$$P(A_n \triangle A_m) \le d_{\mathrm{TV}}(P|_{\mathcal{A}_N}, P^\star|_{\mathcal{A}_N}).$$

Thus the indicators of A_n are Cauchy in P-probability. Their limit has a representative A' belonging to every P-completed $\mathcal{A}_N$. Lemma 2.1 gives $P(A') \in \{0,1\}$. Since

$$|P(A_n) - P^\star(A)| \le d_{\mathrm{TV}}(P|_{\mathcal{A}_n}, P^\star|_{\mathcal{A}_n}) \longrightarrow 0,$$

we obtain $P^\star(A) \in \{0,1\}$, as required. □

For the application to the α-quantum cone, take $\kappa = Q-\alpha > 0$ and add the deterministic radial drift αt. That deterministic addition does not change any of the tail sigma-fields. The proof applies in particular to $d = 3$, without a finite derivative state, a Rice density, or a Markov assertion at a negative cone passage. The proof of the tail theorem itself is identical in every dimension; parity entered only in verifying the common state-forgetting input.

Theorem 2.5 (Decorated exterior zero–one law)**.** *In every dimension $d \ge 3$, let h be the cone in the main theorem and, conditionally on h, let $\mathcal{P} \sim \mathrm{PPP}(M_h)$, with its independent continuous marks. Set*

$$\mathcal{E}_R = \sigma\big(h|_{\mathbb{R}^d\setminus\overline{B}_R}, \mathcal{P}|_{\mathbb{R}^d\setminus\overline{B}_R}\big), \qquad R\in\mathbb{N}.$$

Then $\bigcap_{R\ge1}\overline{\mathcal{E}_R}$ is trivial, with completion under the joint cone and Poisson law.

Proof. The radial component is $S_t^* + \alpha t$ at radius e^{-t}. Its exterior tail is trivial by Theorem 2.4; adding the fixed drift changes none of its sigma-fields.

For the spherical component, smooth compactly supported functions with zero mean on every centered sphere are dense in its Cameron–Martin space. To see this, start with zero-spherical-mean Schwartz functions in the radial–spherical decomposition of [1, Lemma 3.5], then multiply by smooth radial cutoffs. The cutoff approximation converges in the Schwartz topology, hence in the homogeneous Sobolev norm of order $d/2$. It preserves every spherical mean, including in odd dimensions. Gram–Schmidt on a countable dense collection therefore gives an orthonormal Cameron–Martin basis (e_j) whose members each have compact support. In the Gaussian expansion $h_{\mathrm{sph}} = \sum_j \xi_j e_j$ as random distributions, the ξ_j are independent standard normal variables. For each N, restriction outside a sufficiently large ball annihilates $e_1, \dots, e_N$. The spherical exterior tail is therefore contained in the completed independent coefficient tail and is trivial.

The radial and spherical components are independent. For their decreasing exterior sigma-fields, conditional expectation on the product space is the tensor product of the two conditional-expectation projections. Reverse martingale convergence on simple products, followed by L^2 density, shows that their joined decreasing tail is trivial. The exterior sigma-field of their sum is contained in this join, so the exterior tail of h is trivial.

Now let A belong to every completed $\mathcal{E}_R$. Choose measurable representatives at integer radii, simultaneously outside one null set. Given h, restrictions of the marked Poisson process to disjoint integer annuli are independent. Conditional Kolmogorov zero–one therefore gives

$$\mathbf{1}_A = p(h) \quad \text{almost surely,} \qquad p(h) = \mathbb{P}(A \mid h) \in \{0, 1\}.$$

For fixed R, use the representative of A measurable outside $\overline{B}_{R+1}$ and integrate it against the exterior Poisson kernel. The GMC construction is local, so that kernel depends only on the field outside $\overline{B}_R$. Thus $p(h)$ is measurable in every completed field exterior sigma-field. The field tail just proved trivial makes $p(h)$ constant, and consequently $\mathbb{P}(A) \in \{0, 1\}$. □

3 GMC regularity and Voronoi geometry

Lemma 3.1. *There are deterministic $\delta \in (0, 1)$ and $\ell < \infty$ such that almost surely, for every compact $K \subset \mathbb{R}^d$, some random $C_K < \infty$, $c_K > 0$ satisfy*

$$M_h(B(z, r)) + \nu_h(B(z, r)) \le C_K r^{d-\delta}, \qquad M_h(B(z, r)) \ge c_K r^{\ell} \quad (z \in K,\ 0 < r \le 1). \tag{3.1}$$

Both measures have full support and charge no affine hyperplane or Euclidean sphere.

Proof. For an ordinary log-correlated Gaussian field H with bounded covariance remainder near a compact set, the uniform GMC ball estimate is

$$\mu_{H,\beta}(B(z, r)) \le C r^{d+\beta^2/2-\sqrt{2d}\beta-\eta}$$

for each fixed $\eta > 0$; see [1, Lemma A.2] and the moment comparison in [7, 6]. An insertion $-\alpha \log|x|$ costs at most $\alpha\beta$ in the exponent. For balls within distance $2r$ of 0, this follows by summing the ordinary upper bound over dyadic annuli in $B(0, 3r)$; the exponent after insertion is positive. For all other balls bound the insertion weight by a constant times $r^{-\alpha\beta}$. For $\beta \in \{\alpha, \gamma\}$ the resulting loss is at most

$$\sqrt{2d}\gamma + \frac{\gamma^2}{2} + \eta < \frac{1}{2} + \gamma^2 + \eta < 1,$$

using (1.1). The negative-moment and scaling results [7, Propositions 3.6–3.7], with Kahane comparison, give uniformly on compact sets,

$$\mathbb{E}[\mu_{H,\beta}(B(z, r))^{-q}] \le C_q r^{-q(d+\beta^2/2)-\beta^2 q^2/2} \quad (q > 0).$$

Take $q = 1$, a dyadic grid of $O(r^{-d})$ balls, and $\ell > 2d + \gamma^2 + 1$. Markov's inequality and Borel–Cantelli give the lower bound on that grid, and a finer grid and monotonicity give all centers and radii. The insertion weight is bounded below on each compact set. The same reasoning for $\beta = \alpha$ gives full support of ν_h.

These probability-one statements transfer to every cone here by the field local limit in total variation, including $d = 3$. Indeed every rescaled inserted Gaussian field has these local properties,

with random constants; the properties are preserved by every finite dilation and additive constant. On a fixed ball they are measurable from a slightly larger field restriction and its local GMC construction. Total-variation convergence therefore transfers the probability-one event to the cone. A countable exhaustion gives the assertion globally. This avoids any absolute-continuity assertion for the entire three-dimensional Palm path.

A bounded piece of any hyperplane or sphere is covered by $O(r^{-(d-1)})$ balls of radius r. The upper bound gives mass at most $Cr^{1-\delta} \to 0$. The argument is deterministic once the uniform bound holds and hence applies simultaneously to all these surfaces. □

Lemma 3.2. *Almost surely M_h and ν_h have infinite mass in every nonempty open Euclidean cone. The sampled point process is simple and locally finite and has infinitely many points in every such cone. Its Voronoi tessellation is locally finite, all cells are bounded, and its facet-adjacency graph is connected. For each nucleus,*

$$0 < \nu_h(P_v) < \infty, \qquad 0 < \pi(v) < \infty.$$

The origin has a unique nearest nucleus; no $d+1$ nuclei are affinely dependent and no $d+2$ are cospherical.

Proof. The record-Palm radial law satisfies $S_t^* - \kappa t \geq 0$ for $t < 0$. Choose a ball A compactly contained in the intersection of the given cone with $\{1 < |x| < 2\}$ and let $L > 3$. On A, the radial part of $h(L^n \cdot) + Qn \log L$ is at least $-Q \log 2$, since, for $t = -n \log L - \log |x|$,

$$S_t^* + \alpha t + Qn \log L = (S_t^* - \kappa t) - Q \log |x|.$$

The rescaled spherical component has its original law. Hence there are positive fixed-annulus GMC variables $U_{n,\beta}$, with uniformly bounded negative moments of every order, such that

$$M_h(L^n A) \geq c U_{n,\gamma}, \qquad \nu_h(L^n A) \geq c L^{n(d-2)} U_{n,\alpha}.$$

Borel–Cantelli gives $U_{n,\beta} \geq n^{-1}$ eventually, without independence; summing disjoint annuli proves infinite mass. A countable basis of cones gives simultaneity. A cone with another vertex contains a narrower cone with vertex 0 outside a bounded set.

The Poisson claims follow conditionally on the field. Finitely many nuclei in narrow cones covering the directions from v bound its Voronoi cell. If its circumradius is R, all neighbors lie in $B(v, 2R)$, so there are finitely many. The cell contains a ball about its nucleus, giving positive mass and positive finite total conductance. On a compact set the nearest-nucleus distance is finite and Lipschitz, hence bounded; only finitely many nuclei can have cells meeting that set. A compact path between nuclei, perturbed off faces of codimension at least two, crosses finitely many facets and proves connectivity. Finally, the conditional intensity charges no sphere or hyperplane by Lemma 3.1. Successive sampling in bounded regions and exhaustion prove all non-degeneracies and uniqueness at the root. □

4 The finite normalizing constant

We prove (1.3) by finite-volume rooting. No infinite-volume mass-transport identity is used to cancel its reciprocal cell mass.

4.1 Rooting and cancellation

Let $D_0 = B(0, 1/4)$ and let H be the whole-space Gaussian field with unit-sphere average zero. Near $\overline{D}_0$ its covariance is

$$C(x, y) = -\log|x - y| + g(x, y),$$

where g is smooth. Put $M = \mu_{H,\gamma}|_{D_0}$ and $\nu = \mu_{H,\alpha}|_{D_0}$. Write $\mathbb{E}\mu_{H,\beta}(dx) = q_\beta(x)\,dx$, where q_β is smooth, positive and bounded above and below. These deterministic factors account for the chosen spherical-average normalization. Set $Z_0 = \mathbb{E}\nu(D_0) \in (0, \infty)$.

For $m \geq 1$ sample $\mathcal{P}_m \mid H \sim \mathrm{PPP}(mM)$, clip its cells to D_0, and define π_m using only shared interior facets. Under the probability measure

$$\widehat{\mathbb{E}}_m[F] = Z_0^{-1}\mathbb{E}\int_{D_0} F(H, \mathcal{P}_m, z)\nu(dz),$$

let o_m be the nucleus of the cell containing z. Put

$$Y_m = m^{-\vartheta}\frac{\pi_m(o_m)}{\nu(P_{o_m}^{D_0})},$$

with value zero for an empty point process. A single nucleus also gives zero conductance. The partition cells have null boundaries, and integrating the ratio on each cell gives the exact identity

$$\widehat{\mathbb{E}}_m[Y_m] = \frac{m^{-\vartheta}}{Z_0}\mathbb{E}\sum_{v\in\mathcal{P}_m}\pi_m(v). \tag{4.1}$$

Tonelli justifies it before finiteness is known.

4.2 An endpoint GMC integral

Lemma 4.1. *Let G be log-correlated near a compact K, with uniformly bounded covariance remainder and positive one-point normalization factors bounded above and below. Fix $\theta \in (0, 1/8)$ and $\lambda > 0$. Define $D_\gamma = d - 3\gamma^2/2$ and the positive root s by*

$$D_\gamma s + \frac{\gamma^2}{2}s^2 = \lambda.$$

For sufficiently small $r_0 > 0$,

$$\int_0^{r_0} r^{\lambda-1}\sup_{x\in K}\mathbb{E}\exp\{-mr^{-2\gamma^2}\mu_{G,\gamma}(B(x, \theta r))\}\,dr \leq Cm^{-s}, \qquad m \geq 1. \tag{4.2}$$

The constant is uniform under the specified covariance bounds.

Proof. Use exact-variance normalization for Gaussian comparison; the bounded deterministic factors only replace m by a constant multiple. Let W be white noise on $\mathbb{R}^d \times (0, 1]$ with control $dy\,du/(\omega_d u^{d+1})$. The cone field

$$G_c(z) = W\{(y, u) : 0 < u \leq 1,\ |y - z| \leq u\}$$

has covariance $-\log|z - z'| + O(1)$ locally, with the usual small-scale cutoff understood. Add an independent scalar Gaussian of fixed sufficiently large variance. Its covariance dominates that of G on the balls under consideration. Kahane's inequality for the convex function e^{-x} bounds the expectation in (4.2) by this translation-invariant comparison field, uniformly over the center.

Set $r = e^{-t}$. The noise common to every point of $B(0, \theta e^{-t})$ contains

$$J_t = \{(y, u) : 2\theta e^{-t} \le u \le 1,\ |y| \le u - \theta e^{-t}\}.$$

These sets increase with t, so $W(J_t) = B^0_{V(t)}$ for a standard Brownian motion, where

$$V(t) = \int_{2\theta e^{-t}}^1 \left(1 - \frac{\theta e^{-t}}{u}\right)^d \frac{du}{u}, \qquad V'(t) = (1 - \theta e^{-t})^d, \quad V(t) = t + O(1).$$

Remove this common noise and rescale the ball. The mass has the exact form

$$e^{2\gamma^2 t}\mu_{G_c,\gamma}(B(0, \theta e^{-t})) = e^{\gamma B^0_{V(t)} - D_\gamma V(t)} W_t.$$

For each t, $W_t > 0$ is independent of $B^0_{V(t)}$ and includes the added scalar Gaussian and the bounded factor $\exp\{(d - 2\gamma^2)(V(t) - t)\}$. Its residual field covariance, after rescaling, is $-\log|\xi - \xi'| + O(1)$ uniformly in t. Thus the all-dimensional inverse-moment theorem [7, Proposition 3.6], with Kahane comparison, give $\sup_{t \ge t_0} \mathbb{E} W_t^{-q} < \infty$ for every $q > 0$.

Choose $q > s$. The inequality $\mathbb{E} e^{-uW_t} \le C_q(1 \wedge u^{-q})$ and the change of variable $v = V(t)$ bound the left side of (4.2) by

$$C \int_0^\infty e^{-\lambda v} \mathbb{E}[1 \wedge (cme^{\gamma B^0_v - D_\gamma v})^{-q}]\, dv.$$

The λ-resolvent density of $\gamma B^0_v - D_\gamma v$ from zero is

$$u_\lambda(y) = \frac{1}{k_\lambda} \begin{cases} e^{sy}, & y \le 0, \\ e^{-s_+ y}, & y \ge 0, \end{cases} \quad k_\lambda = \sqrt{D_\gamma^2 + 2\lambda\gamma^2}, \quad s = \frac{k_\lambda - D_\gamma}{\gamma^2}, \quad s_+ = \frac{k_\lambda + D_\gamma}{\gamma^2}.$$

This follows by integrating the drifted Gaussian transition density. Integrate $(1 \wedge (cme^y)^{-q})u_\lambda(y)$, splitting at $-\log(cm)$ and at zero. For $cm \ge 1$ the three terms are bounded respectively by Cm^{-s}, Cm^{-s} and Cm^{-q} since $q > s$. The bounded remaining range of m is absorbed into C. □

4.3 Total conductance

For distinct $v, w \in D_0$, put $H_{v,w} = \{x : |x - v| = |x - w|\}$, $R(v, w, x) = |x - v|$, and

$$M^{v,w}(dz) = e^{\gamma^2 C(v,z) + \gamma^2 C(w,z)} M(dz).$$

The ordered two-point Poisson formula and the Gaussian insertion identity [8, Theorem 17] give

$$\mathbb{E} \sum_v \pi_m(v) = m^2 \int_{D_0^2} \frac{q_\gamma(v) q_\gamma(w) e^{\gamma^2 C(v,w)}}{|v - w|} \cdot \int_{H_{v,w} \cap D_0} \mathbb{E} e^{-mM^{v,w}(B(x, R(v,w,x)) \cap D_0)}\, d\mathcal{H}^{d-1}(x)\, dv\, dw. \tag{4.3}$$

Indeed x belongs to the common facet precisely when its open ball to v, w contains no other nucleus, except for lower-dimensional intersections with zero facet measure. Gaussian insertion supplies the two displayed weights. Localization away from diagonals, followed by Tonelli, makes the calculation valid without prior integrability. The ordered sum counts each edge twice on both sides.

Proposition 4.2. *$\mathbb{E} \sum_{v \in \mathcal{P}_m} \pi_m(v) \le Cm^\vartheta$ for all $m \ge 1$.*

Proof. For $R < r_0$, the ball D_0 has a uniform interior-ball property: some $B(y_0, \theta R)$ lies in $B(x, R/4) \cap D_0$, for a fixed $\theta > 0$. Both insertion distances are comparable to R there, giving

$$M^{v,w}(B(x,R) \cap D_0) \geq cR^{-2\gamma^2} M(B(y_0, \theta R)).$$

Write $v = x - y - ue$, $w = x - y + ue$, where $e \in S^{d-1}$, $u > 0$, and $y \in e^{\perp}$. Midpoint and difference coordinates give, up to a dimension-dependent factor,

$$\frac{dv\, dw\, d\mathcal{H}^{d-1}(x)}{|v-w|} = C_d u^{d-2}\, dx\, de\, du\, dy.$$

The remaining covariance weight is at most $Cu^{-\gamma^2}$. With $R = (u^2 + |y|^2)^{1/2}$ and $a = u/R$,

$$u^{d-2-\gamma^2}\, du\, dy = C_d R^{2d-3-\gamma^2} a^{d-2-\gamma^2} (1-a^2)^{(d-3)/2}\, dR\, da\, d\zeta.$$

The angular integral is finite since $\gamma^2 < d-1$. Lemma 4.1 with $\lambda = 2d - 2 - \gamma^2$ has exponent $s = 2 - \vartheta$, because (1.2) gives

$$\left(d - \frac{3\gamma^2}{2}\right)(2-\vartheta) + \frac{\gamma^2}{2}(2-\vartheta)^2 = 2d - 2 - \gamma^2.$$

Thus the small-R contribution in (4.3) is $O(m^{2-s}) = O(m^{\vartheta})$.

For $R \geq r_0$, $B(x,R) \cap D_0$ contains a ball of fixed positive radius uniformly in $x \in D_0$. Insertion weights are bounded below on D_0, so negative moments give a void probability $O(m^{-N})$ for every $N > 0$. Bisector sections have bounded area and $\int_{D_0^2} |v-w|^{-1-\gamma^2}\, dv\, dw < \infty$. Choose $N > 2$ to bound this contribution uniformly. □

4.4 Passage to the cone

Under the rooted law, z has density $q_\alpha(z)/Z_0$. Conditional on z, insertion replaces H by

$$H^z = H + \alpha C(z, \cdot) = H - \alpha \log|\cdot - z| + f_z, \qquad f_z(x) = \alpha g(z,x).$$

Set $b_m = \gamma^{-1} \log m$. Apply the physical first-passage scale to $H - \alpha \log|\cdot - z|$ at level $b_m + f_z(z)$, calling it $r_{m,z}$. Here H keeps its original remote normalization. Corollary 5.7 justifies this version of the local limit: it uses local Gaussian equivalence and the trivial shrinking germ, rather than substituting the random variable $H_1(z)$ into a deterministic-level theorem. For fixed interior z, this scale tends to zero. The rescaled field

$$h_{m,z}(x) = H^z(z + r_{m,z}x) + Q \log r_{m,z} + b_m$$

equals the first-passage field plus $f_z(z + r_{m,z}x) - f_z(z)$, which tends uniformly to zero on each fixed ball. The local field total-variation limit, followed by local measurable GMC and this smooth multiplier, gives joint local convergence of the field and both GMC measures to those of h.

Let $D_{m,z} = r_{m,z}^{-1}(D_0 - z)$ and $\widetilde{\mathcal{P}}_{m,z} = r_{m,z}^{-1}(\mathcal{P}_m - z)$. Exact covariance gives

$$\widetilde{\mathcal{P}}_{m,z} \mid (H^z, r_{m,z}) \sim \mathrm{PPP}(\mu_{h_{m,z},\gamma}|_{D_{m,z}}), \qquad \mu_{h_{m,z},\alpha}(A) = m^{\vartheta} r_{m,z}^{2-d} \mu_{H^z,\alpha}(z + r_{m,z}A).$$

Conductances acquire the factor $r_{m,z}^{2-d}$, so Y_m is exactly the conductance-to-mass ratio of the rescaled clipped root cell.

Lemma 4.3. *For locally convergent simple point configurations whose limit has a bounded root cell, a unique nearest nucleus to zero and no affine or cospherical degeneracies, the root-cell conductances converge. If measures also converge locally weakly, and the limiting root cell has positive mass and null boundary, its conductance-to-mass ratio converges. Clipping to domains exhausting $\mathbb{R}^d$ has the same conclusion.*

Proof. Finitely many Voronoi half-spaces determine and bound the limiting cell and a fixed neighborhood of it. A sufficiently large ball contains those nuclei and every possible neighbor; nuclei outside a still larger ball are too far to affect that neighborhood. These strict inequalities persist under local perturbations. Non-degeneracy makes the resulting finite diagram stable, with Hausdorff convergence of cells and convergence of facet areas and conductances. For the cell mass, if $P_n \to P$, the finite half-space descriptions imply $P^{-\eta} \subset P_n \subset P^{+\eta}$ eventually. Portmanteau on the open inner and compact outer sets, followed by $\eta \downarrow 0$, proves $\mu_n(P_n) \to \mu(P) > 0$. Exhausting domains eventually contain the shielding neighborhood. □

Proposition 4.4. *The constant Z in (1.3) is finite and strictly positive in every dimension under consideration.*

Proof. Local weak convergence of the intensity measures implies joint convergence of their Poisson samples, since their conditional Laplace functional is $\exp\{-\int(1-e^{-f})\,dM\}$ for $f \in C_c^+(\mathbb{R}^d)$. For fixed interior z, the domains $D_{m,z}$ exhaust space. Lemmas 3.2 and 4.3, with localization to a large shielding ball, therefore imply

$$Y_m \implies \frac{\pi(o)}{\nu_h(P_o)}.$$

The fixed integrable density q_α/Z_0 allows mixing over z by dominated convergence. By (4.1) and Proposition 4.2, $\sup_m \widehat{\mathbb{E}}_m Y_m < \infty$. Apply bounded-continuous convergence to $Y_m \wedge A$ and then let $A \uparrow \infty$. This proves finiteness without uniform integrability. Strict positivity follows from Lemma 3.2. □

5 Liouville Brownian motion and rerooting

This section proves rerooting by one local-limit argument in every dimension, following the strategy of [1, Section 6.4]. Throughout, $d \geq 3$, $Q > \sqrt{2d}$, and $\alpha = Q - \sqrt{Q^2-4}$. Thus

$$\alpha Q - \frac{\alpha^2}{2} = 2, \qquad 0 < \alpha < 2. \tag{5.1}$$

The field h is the α-cone, with the radial and spherical normalizations used above. We use the already constructed cone local-limit theorem, the ordinary radial height-and-memory comparison, and the first-passage scaling law in Proposition 2.3. In even dimensions the corresponding radial and cone inputs are those of [1]; in odd dimensions they are the Volterra and record-Palm results proved above. The diffusion convention is Brownian generator $\frac{1}{2}\Delta$.

5.1 The clock at the cone point

Lemma 5.1 (Finite initial clock and infinite lifetime)**.** *The α-GMC Brownian clock of the cone is finite on bounded Brownian time intervals, continuous, and strictly increasing when started at 0. Its total lifetime is infinite almost surely. In particular, the associated two-sided LBM $X = (X_t)_{t\in\mathbb{R}}$, obtained from conditionally independent Brownian branches at 0, is well defined and continuous for all times.*

Proof. Write $h_{\mathrm{rad}}(e^{-s}\theta) = S_s^* + \alpha s$. The record-Palm identification gives $S_s^* = o(|s|)$ as $s \to \pm\infty$: ordinary radial sublinearity is invariant under every finite recentering, and consequently holds at record-Palm-almost every recentering. The spherical field is independent of S^*. Its GMC normalization and dilation invariance give

$$\mathbb{E}[\mu_{h_{\mathrm{sph}},\alpha}(B(0,e^{-k}))] = c_\alpha e^{-k(d+\alpha^2/2)}, \qquad k \geq 0,$$

where $c_\alpha < \infty$. The finiteness is also obtained by integrating the one-point regularization expectation on the unit ball. Conditioning on S^* and summing over annuli therefore gives

$$\mathbb{E}\left[\int_{B(0,1)} |x|^{2-d}\nu_h(dx)\middle| S^*\right] \leq C\sum_{k\geq 0}\exp\left\{-k(2-\alpha^2/2) + \alpha \sup_{k\leq s\leq k+1} S_s^*\right\} < \infty.$$

In particular the Brownian Green potential at 0 is finite. Away from 0, couple h to a whole-space LGF using the same spherical component. Their difference is a continuous radial function on every compact annulus. The strict ordinary-LGF clock construction consequently applies there, with multiplication by a positive continuous density. The Green-potential bound supplies the finite clock at the initial point; Brownian motion in $d \geq 3$ does not return to this point after time zero. Continuity and strict increase follow from the local clock construction and its full support. This argument uses no radial derivative and includes $d = 3$.

It remains to check lifetime, rather than infer it from local finiteness. Let $T_h \in (0,\infty]$ be the full Brownian clock of one branch. For any fixed $b > 0$, Proposition 2.3 gives a field-measurable $r_b > 0$ for which

$$h(r_b\,\cdot) + Q\log r_b + b \overset{\text{law}}{=} h.$$

Brownian scaling, conditional on the field, and (5.1) imply $T_h \overset{\text{law}}{=} e^{\alpha b}T_h$. A probability distribution on $(0,\infty]$ with this invariance is concentrated at infinity. Thus $T_h = \infty$ almost surely. □

Lemma 5.2 (The inserted Gaussian reference clock)**.** *For the whole-space LGF H with $H_1(0) = 0$, the field $G = H - \alpha\log|\cdot|$ defines a continuous two-sided LBM path for all physical times, with finite initial clock and infinite lifetime on both branches.*

Proof. Its finite initial clock follows from the same Green-potential annular estimate as above, with the ordinary sublinear radial process S in place of S^*. On compact annuli its clock is the ordinary LGF clock multiplied by the positive continuous density $|x|^{-\alpha^2}$. For $R > 1$, put

$$H^R = H(R\,\cdot) - H_R(0), \qquad c_R = H_R(0) + (Q-\alpha)\log R.$$

The normalized field H^R has the law of H, and two-sided radial sublinearity gives $c_R \to \infty$ almost surely. If τ_R is the Brownian exit time from $B(0,R)$, then Brownian scaling and clock covariance give the exact identity

$$F_G(\tau_R) = e^{\alpha c_R}V_R,$$

where V_R is the unit-ball clock for $H^R - \alpha\log|\cdot|$ along the Brownian path $R^{-1}B_{R^2\cdot}$. Each V_R has the same law as a strictly positive finite random variable V. No independence of V_R and c_R is asserted or needed. For $M, \delta > 0$,

$$\mathbb{P}(F_G(\tau_R) \leq M) \leq \mathbb{P}(e^{\alpha c_R} \leq M/\delta) + \mathbb{P}(V \leq \delta).$$

First let $R \to \infty$, then $\delta \downarrow 0$. Since $F_G(\tau_R)$ is nondecreasing in R, its divergence in probability implies almost-sure divergence. Both Brownian branches therefore define continuous reference LBM paths for all physical times. □

5.2 The joint local limit and stability under Palm densities

For a whole-space LGF H, normalized by $H_1(0) = 0$, let $\bar{H} = H - \alpha \log|\cdot|$. Put

$$\sigma_b = \inf\{s \in \mathbb{R} : H_{e^{-s}}(0) - (Q-\alpha)s = -b\}, \qquad H_b = \bar{H}(e^{-\sigma_b}\,\cdot) - Q\sigma_b + b.$$

All first passages here are global first passages, not restarts of the conditioned cone at a negative time.

Lemma 5.3 (Field, chaos, and stopped-path local limit)**.** *For each fixed $R < \infty$, the field H_b on $B(0,R)$, its GMC measures, and its two-sided LBM stopped on exiting $B(0,R)$ converge jointly in total variation to the corresponding cone objects. Consequently*

$$\left(H_b, e^{\sigma_b} X^{\bar{H}}_{e^{-\alpha b}\cdot}\right) \implies (h, X)$$

locally in the distribution topology for the field and the uniform topology for the two-sided paths.

Proof. The cone theorem gives total variation convergence of the field restrictions, not just weak convergence of distributions. Given the field, the process $e^{\sigma_b} B_{e^{-2\sigma_b}\cdot}$ is an independent two-sided standard Brownian motion: its conditional law does not depend on the field-measurable scale. Hence taking its product with the field law preserves total variation distance. Locally defined GMC and the stopped clock are measurable functions of this pair. Use a slightly larger open ball to define mollifiers, and then stop in $B(0,R)$; data processing gives the claimed total variation convergence. This does not use continuity of GMC as a function of a distribution.

Equation (5.1) gives the exact identity

$$X^{H_b}_t = e^{\sigma_b} X^{\bar{H}}_{e^{-\alpha b}t}.$$

Lemma 5.2 supplies finite initial clocks and conservativity for the inserted reference field. The limit is conservative by Lemma 5.1. Stopped-path convergence followed by localization at $R \to \infty$ therefore gives the asserted unrestricted local path convergence. □

Lemma 5.4 (Trivial shrinking germ)**.** *For an ordinary whole-space LGF H and an independent two-sided Brownian motion rooted at 0, the completed germ generated by $H|_{B(0,r)}$ and the two Brownian branches up to their first exits from $B(0,r)$, as $r \downarrow 0$, is trivial. The same holds after adding the deterministic insertion $-\alpha \log|\cdot|$.*

Proof. Decompose H into its independent radial and spherical components. The height-and-memory comparison for the unconditioned radial process implies triviality of its completed future tail. Explicitly, for each fixed Brownian past, the conditional law of the entire radial path after time n approaches the unconditional law in total variation. Conditional expectations of any tail-event representative therefore equal its unconditional probability; martingale convergence as the fixed past increases proves the assertion. The height term must be included in this comparison, as in the radial-tail argument above.

The spherical Cameron–Martin space has a countable dense family of smooth zero-spherical-mean functions supported in compact annuli. To verify the inner cutoff, a smooth zero-spherical-mean function f satisfies $f(0) = 0$. For a fixed smooth cutoff χ near 0, critical Sobolev scaling and Taylor's theorem give

$$\|\chi(\cdot/r) f\|_{\dot{H}^{d/2}} = O(r).$$

Indeed $r^{-1} f(r\,\cdot)\chi$ stays bounded in every fixed compactly supported Sobolev norm, and the $\dot{H}^{d/2}$ norm is dilation invariant. Outer cutoffs follow from Schwartz decay. Gram–Schmidt thus gives an

annularly supported orthonormal Cameron–Martin basis (e_j). In the Gaussian expansion $H_{\mathrm{sph}} = \sum_j \xi_j e_j$, each sufficiently small ball sees only $(\xi_j)_{j>N}$. The spherical inner germ is trivial by Kolmogorov's zero–one law.

Blumenthal's zero–one law handles the Brownian germ. For independent factors, reverse conditional expectation projections on the product L^2 space are tensor products; their limits show that the join of these decreasing germs is trivial, including completion. A deterministic insertion changes none of their sigma-fields. □

Lemma 5.5 (Stable local zoom)**.** *The local joint limit of Lemma 5.3 is unchanged by any fixed probability-law change absolutely continuous with respect to the reference inserted-field/Brownian law on a neighborhood of the root, when the zoom uses the first passage inside that neighborhood. The same conclusion holds for a global first passage if, under the changed law, it eventually agrees with the neighborhood-restricted passage and tends to the root. These locality hypotheses are also required after localizing to a random neighborhood.*

Proof. Here is the precise density argument. Write $\mathcal{G}_n$ for the shrinking-neighborhood sigma-fields of Lemma 5.4. For any bounded test of a fixed stopped zoom window, replace the global first passage by the first passage after n. The replacement agrees with the original under the reference law with probability tending to one as $b \to \infty$: sublinearity makes the minimum before n finite, and $\sigma_b \to \infty$. Under a merely locally absolutely continuous law, we first work with its neighborhood-restricted passage. For a global passage, the stated additional locality assumption supplies agreement; it does not follow from local absolute continuity alone. For the same reason every fixed rescaled spatial window is eventually inside $B(0, e^{-n})$. The replaced stopped zoom is consequently $\mathcal{G}_n$-measurable, up to an event of vanishing probability. More explicitly, for a radius-R stopped window use a fixed fallback value unless its passage satisfies $\sigma_b > n + \log R$; this failure probability tends to zero. On the complement both the required field restriction and the Brownian path up to the scaled exit are available from $\mathcal{G}_n$.

If $D \geq 0$ is a density with $\mathbb{E}D = 1$ and $|f| \leq 1$, the resulting change in $\mathbb{E}[Df]$ tends to zero for fixed n, by integrability of D. For the measurable replacement $Y_b^{(n)}$,

$$\left|\mathbb{E}[Df(Y_b^{(n)})] - \mathbb{E}[f(Y_b^{(n)})]\right| \leq \mathbb{E}\left|\mathbb{E}[D \mid \mathcal{G}_n] - 1\right|.$$

Reverse martingale convergence and germ triviality make this bound tend to zero as $n \to \infty$. This proves stability, uniformly over bounded tests, for stopped windows. Lemma 5.1 and localization then give the full local path limit. A density given only on a fixed neighborhood is handled there, and increasing localization events cover random neighborhoods. □

Lemma 5.6 (Local equivalence of remote normalizations)**.** *Two whole-space LGF normalizations whose averaging distributions are supported outside an open ball U have equivalent laws on each ball $U' \Subset U$. The same is true after adding a deterministic function smooth on U. In particular this applies to the original unit-sphere normalization and the unit-sphere normalization centered at a deterministic point z when U is a sufficiently small ball around z disjoint from both normalization spheres.*

Proof. Let $\mathcal{V}_U$ be the closure of $C_c^\infty(U)$ in the homogeneous LGF Cameron–Martin space $\dot{H}^{d/2}$. Project the underlying isonormal Gaussian onto $\mathcal{V}_U$, using the decomposition of [10, Proposition 5.1]. Its field component H_U is the same for both normalizations: the remote averaging functionals vanish on every element of this subspace. In either normalization the independent orthogonal remainder J satisfies

$$(-\Delta)^{d/2} J = 0 \quad \text{in } U$$

in the distributional sense, by orthogonality against $C_c^\infty(U)$. The all-dimensional regularity theorem [10, Theorem 8.3] makes J smooth on U. This argument does not replace the fractional operator by a local differential operator in odd dimensions.

Choose $\chi \in C_c^\infty(U)$ equal to one on U'. For each realization of J, $\chi J \in \mathcal{V}_U$. Conditional on J, the restricted field $(H_U + J)|_{U'}$ is thus a Cameron–Martin translate of $H_U|_{U'}$ and has an equivalent law. Integrating these conditional equivalent laws preserves their null sets. This proves equivalence for the two normalizations, without an unproved nondegeneracy assertion for a finite-rank covariance perturbation. The deterministic smooth addition is treated with the same cutoff. □

Corollary 5.7 (Physical levels and remote normalizations)**.** *Let z be deterministic, and suppose a field G on a neighborhood of z has law locally absolutely continuous with respect to $H(\cdot - z) - \alpha \log|\cdot - z|$, where $H_1(0) = 0$. Assume its physical global passage*

$$\sigma_b(G, z) = \inf\{s \in \mathbb{R} : G_{e^{-s}}(z) - Qs = -b\}$$

is finite, tends to infinity, and agrees eventually with the passage restricted to $s \geq n$, for each fixed n. Then the field

$$G(z + e^{-\sigma_b(G,z)}\,\cdot) - Q\sigma_b(G, z) + b$$

has the cone local limit at the deterministic physical level b. The conclusion holds jointly with an independent Brownian clock, or with a Brownian pair whose joint local law has the preceding absolute continuity property.

Proof. Apply Lemma 5.5 to the local law of the raw field G, using the same deterministic-level passage functional on both laws. The passage assumptions ensure the requisite germ measurability. In particular, by Lemma 5.6, changing a remote Gaussian normalization, or adding a smooth deterministic covariance remainder from a GMC insertion, can be handled by local Gaussian equivalence and Cameron–Martin translation. One does not substitute a correlated random number such as $H_1(z)$ into a deterministic-level convergence theorem. The statement permits such a normalization only when its resulting raw local field law has the specified absolute continuity. We verify the passage hypothesis explicitly for the raw fields used here. For deterministic z, write

$$S_s^z = H_{e^{-s}}(z) - H_1(z).$$

This process has the ordinary radial law and is sublinear in both directions; independence from $H_1(z)$ is unnecessary. Thus for $G = H - \alpha \log|\cdot - z|$,

$$G_{e^{-s}}(z) - Qs = S_s^z + H_1(z) - (Q - \alpha)s \longrightarrow \begin{cases} +\infty, & s \to -\infty, \\ -\infty, & s \to +\infty. \end{cases}$$

For the actual Campbell insertion, let σ be normalized surface measure on the original normalization sphere and set

$$\ell(w) = \int \log|w - \theta|\,\sigma(d\theta), \qquad c_0 = \iint \log|\theta - \eta|\,\sigma(d\theta)\sigma(d\eta).$$

The normalized covariance is

$$C(w, z) = -\log|w - z| + \ell(w) + \ell(z) - c_0.$$

For z off that sphere, ℓ is smooth near z, so the average of $C(\cdot, z)$ on the radius-e^{-s} sphere about z equals $s + O_z(1)$ as $s \to +\infty$. At infinity $\ell(w) = \log|w| + O(|w|^{-1})$, and hence $C(w, z) = O_z(1)$ as $|w| \to \infty$. Consequently the Campbell field $G = H + \alpha C(\cdot, z)$ satisfies

$$G_{e^{-s}}(z) - Qs = \begin{cases} -Qs + o(|s|), & s \to -\infty, \\ -(Q-\alpha)s + o(s), & s \to +\infty. \end{cases}$$

It therefore has the same positive and negative limiting signs. Continuity of these radial averages implies that their minimum on every half-line $(-\infty, n]$ is finite. Their first global hit of $-b$ consequently exists, tends to $+\infty$, and, for every fixed n, eventually equals the first hit restricted to $s \geq n$. This controls all exterior scales, rather than only a bounded scale interval. A deterministic additive normalization constant changes neither conclusion. □

5.3 Rerooting at an LBM time

Theorem 5.8 (Cone rerooting in every dimension $d \geq 3$)**.** *For each deterministic $t \in \mathbb{R}$, there is a random $R_t \in (0, \infty)$, possibly realized on an extension of the probability space, such that*

$$\Big(h(X_t + R_t\,\cdot) + Q \log R_t, R_t^{-1}(X_{t+\cdot} - X_t)\Big) \overset{\text{law}}{=} (h, X). \tag{5.2}$$

In particular the field and path law is stationary under time shifts and spatial rerooting modulo dilation. If a conditionally Poisson decoration with intensity M_h is retained, the same conclusion holds with that decoration transformed by the spatial map.

Proof. Start with an ordinary whole-space LGF and an independent two-sided Brownian motion from 0. Consider the LBM time interval corresponding to Brownian times $[-1, 1]$, whose finite positive length is L. Given the field and Brownian path, sample U uniformly on this LBM interval.

We spell out the rooting input. Here $0 < \mathbb{E}L < \infty$: the one-point GMC regularization expectation for the unit-sphere-normalized LGF has at most polynomial growth in the spatial location, which is integrable at Brownian times in $[-1, 1]$. The finite Campbell probability is $(\mathbb{E}L)^{-1} dF$ times the original joint field/Brownian law. Replacing normalized sampling dF/L by this probability biases the joint law by $L/\mathbb{E}L$; the reverse density is exactly $(\mathbb{E}L)/L$. The GMC insertion identity [8, Theorem 17], applied conditionally on the Brownian path, says that Gaussian exponential tilting at Brownian time s inserts $\alpha C(\cdot, B_s)$ into the field. Its singular part near B_s is $-\alpha \log|\cdot - B_s|$. The remainder is smooth on a sufficiently small neighborhood, outside a null set where the root lies on the normalization sphere. The local equivalence in Lemma 5.6, applied before adding the common logarithmic insertion, gives local absolute continuity with respect to the reference inserted LGF. Corollary 5.7 retains the physical passage level under this change of normalization.

Conditionally on s and B_s, the Brownian branches adjacent to s are locally absolutely continuous with respect to two independent Brownian motions from B_s. The past branch is a Brownian bridge on the side containing the original time zero; its density on a shorter interval is the usual positive heat-kernel ratio. Localize away from $s = 0, \pm 1$ and then exhaust; those times have zero Campbell mass. The other branch has the Brownian Markov law. Gaussian tilting is independent of the remaining Brownian path given the rooted location. Thus the joint rooted field/Brownian law is locally absolutely continuous with respect to the reference pair. Finally the density proportional to $1/L$ is retained, not discarded. This is the Gaussian Campbell identity for the Brownian occupation chaos; all its operations are independent of parity. The finite Campbell probability just specified makes the density change a probability-law change without a formal infinite-mass normalization.

The global-passage requirement is also satisfied under this rooted law: conditional on the Brownian-time root location z, its Gaussian field is precisely the Campbell insertion just analyzed. The Campbell distribution of z is absolutely continuous with respect to Lebesgue measure, so the original normalization sphere has zero root probability. Integrating the deterministic-z almost-sure passage conclusion gives the required assertion under the finite Campbell probability. Multiplication by the retained integrable reverse density $(\mathbb{E}L)/L$ preserves this probability-one assertion. Thus the use of a global physical passage in the stable-zoom lemma has been checked, in addition to its local absolute-continuity hypothesis.

For $b > 0$ let A_b be the field/path pair zoomed at LBM time U, with the deep first-passage normalization at level $-b$ and the additive constant b. Lemmas 5.3 and 5.5 give $A_b \Longrightarrow (h, X)$. For fixed t, let B_b be the analogous zoom at $U + e^{-\alpha b}t$. Given the original field/path pair, the total variation distance between these two choices of root time is at most

$$\min\{1, e^{-\alpha b}|t|/L\}.$$

It tends to zero in expectation. Therefore $B_b \Longrightarrow (h, X)$ as well, with the same LBM-time parametrization.

It is important not to assert continuity of a global first-passage normalization map. The following argument avoids that assertion. Write $A_b = (a_b, x_b)$ and $B_b = (b_b, y_b)$, and let R_b be the ratio of their spatial scales. They satisfy exactly

$$b_b = a_b(x_b(t) + R_b\,\cdot) + Q \log R_b, \qquad y_b = R_b^{-1}(x_b(t + \,\cdot) - x_b(t)).$$

For $t \neq 0$, evaluating at $-t$ gives

$$R_b = \frac{|x_b(t)|}{|y_b(-t)|}.$$

The limiting LBM has no visit to 0 at a nonzero time: the clock is strict with infinite lifetime, and a Brownian point is polar in $d \geq 3$. Both the numerator and denominator thus have tight, strictly positive limits in distribution. Consequently (R_b) is tight in $(0, \infty)$. Take a joint subsequential limit of (A_b, B_b, R_b). Translation and dilation act continuously on distributions and continuous paths when the dilation stays in a compact subset of $(0, \infty)$. The exact identities therefore pass to the limit, while both pair marginals have law (h, X). This proves (5.2). For $t = 0$, use $R_t = 1$.

The limiting dilation can be sampled conditionally on the field/path pair, before sampling any Poisson decoration. Conditional on these objects, Poisson mapping and GMC covariance identify the transformed intensity as that of the transformed field. That intensity depends only on the transformed field, so forgetting the original representation preserves the claimed decorated joint law. □

Remark 5.9 (Scope and inputs). The theorem extends the rerooting statement, not by declaring the even-dimensional finite-state radial process available in odd dimensions, but by using the odd cone total-variation theorem and the unconditioned Volterra forgetting estimate. The argument uses the standard subcritical GMC Campbell identity and strict Brownian PCAF construction; it does not give an independent reconstruction of those theories. For the quenched walk theorem only the smaller parameter range stated there is needed. The path in (5.2) is explicitly shifted by t; omitting this time shift would give a false joint field/path statement.

6 Stationarity and ergodicity of the environment

Throughout this section $d \geq 3$. Write Q_0 for the law of the cone together with its unit-intensity marked Poisson process, with continuum root 0. The marks give a measurable allocation of every

Voronoi tie point. Let $c_\omega(x)$ be the allocated nucleus at x, and put $m(v) = \nu_h(P_v)$. The law $\mathsf{Q}_{\mathrm{cell}}$ is obtained by choosing $o = c_\omega(0)$, rooting the decorated environment at o, and forgetting the continuum mark. Vertex-rooted and continuum-rooted environments are viewed modulo positive dilations; a translation θ_z moves the root to z before this quotient is taken.

We use the following previously established properties. Lemma 3.1 gives, on each compact set, the uniform bound $\nu_h(B(x,r)) \le Cr^{d-\delta}$ for some $\delta < 1$. The strict full-support PCAF comes from the ordinary-GMC clock on each annulus, multiplied by the positive continuous radial density of the cone, together with Lemma 5.1 at 0. Topological full support of a measure alone is not being used to infer strictness of its PCAF. The all-start construction below identifies these local clocks. The cone diffusion from 0 is conservative, and Theorem 5.8 gives invariance of its two-sided law under LBM-time shifts followed by rerooting and a positive dilation. Its positive and negative branches are conditionally independent copies started at 0. The Poisson geometry gives bounded, locally finite cells, a connected facet graph, positive finite incident conductances, and infinitely many nuclei in each nonempty open Euclidean cone. These geometric properties, rather than stationarity alone, will be used below. Proposition 4.4 gives

$$0 < Z = \mathbb{E}_{\mathsf{Q}_{\mathrm{cell}}}\left[\frac{\pi(o)}{m(o)}\right] < \infty. \tag{6.1}$$

6.1 A symmetric resolvent at every starting point

The point 0 has zero ν_h-mass. Consequently, symmetry of a resolvent density only $\nu_h \otimes \nu_h$-almost everywhere would not justify exchanging this fixed root with a typical LBM location. We first construct a version with no exceptional starting points.

Lemma 6.1 (All-start symmetric resolvent)**.** *For almost every field, there is a jointly measurable function $r_h(x,y) \in (0,\infty]$, defined for $x \neq y$, such that*

$$\mathbb{E}^h_x\left[\int_0^\infty e^{-t} f(X^h_t)\,dt\right] = \int r_h(x,y) f(y)\,\nu_h(dy), \qquad x \in \mathbb{R}^d, \tag{6.2}$$

for every nonnegative Borel function f. It satisfies $r_h(x,y) = r_h(y,x) > 0$ for every $x \neq y$ and is finite for ν_h-almost every y in each fixed row x. If $h_{z,r} = h(z + r\,\cdot) + Q\log r$, then

$$r_{h_{z,r}}(r^{-1}(x-z), r^{-1}(y-z)) = r^{d-2} r_h(x,y). \tag{6.3}$$

Proof. Fix a field satisfying the local mass bound. Restrict its measure to a compact neighborhood of a ball D, calling the resulting measure μ. For the standard Brownian heat kernel p_s, the ball bound and Gaussian decay imply

$$\sup_x \int_0^t \int p_s(x,y)\,\mu(dy)\,ds \le Ct^{1-\delta/2}, \qquad 0 < t \le 1. \tag{6.4}$$

In particular μ is a Kato measure. Smooth it by convolution to get smooth densities μ_j with the same estimate uniformly in j. Their potentials converge uniformly in the starting point and in the time horizon on each compact time interval. Indeed, the small-time part is uniformly small by (6.4); on times bounded away from zero the heat kernels form an equicontinuous family, and weak convergence of the finite measures applies.

Here are details ensuring convergence of clocks from every starting point. Put $U_t\rho(x) = \int_0^t \int p_s(x,y)\rho(dy)\,ds$ and $A^j_t = \int_0^t \mu_j(B_s)\,ds$. The Brownian Markov property gives

$$\mathbb{E}_x[(A^i_t - A^j_t)^2] \le 2 \sup_{s \le T} \|U_s(\mu_i - \mu_j)\|_\infty \|U_T(\mu_i + \mu_j)\|_\infty, \qquad t \le T.$$

The time-ordered moment formula and (6.4) also give $\mathbb{E}_x|A_t^j - A_s^j|^k \le k! C^k |t-s|^{k(1-\delta/2)}$, uniformly in x, j on a fixed time interval. Choose k so that the exponent exceeds one. The resulting uniform modulus bound and the finite-time L^2 estimate imply locally uniform convergence in probability of the clocks, uniformly over starting points. The limit has Revuz measure μ, so agrees with the prescribed PCAF on paths stopped in D. This agreement holds from every start: after any deterministic positive Brownian time, the position has a Lebesgue density and avoids the polar exceptional set in PCAF uniqueness. Additivity gives agreement of the subsequent increments; letting rational positive times decrease to zero and using continuity removes the exceptional initial point. The same construction applies to weighted measures dominated by a constant times μ.

Consider a Brownian bridge from x to y in duration $s > 0$. Its first half is absolutely continuous with respect to Brownian motion from x, with density $p_{s/2}(B_{s/2}, y)/p_s(x, y)$; its reversed second half has the analogous density from y. These densities are bounded uniformly when the endpoints stay in a compact set and s stays in a compact subset of $(0, \infty)$. The preceding smooth-clock construction therefore defines a finite bridge clock $F_h(s)$ for every pair of endpoints. It is invariant under bridge reversal. Define

$$r_h^D(x,y) = \int_0^\infty p_s(x,y)\mathbb{E}^{x,y;s}\Big[e^{-F_h(s)}\mathbf{1}_{\{B[0,s]\subset D\}}\Big]\,ds, \qquad x,y\in D, \quad x\neq y. \tag{6.5}$$

It is symmetric, strictly positive, and bounded above by the killed Brownian Green function $G_D(x, y)$.

For each smooth measure, Brownian occupation followed by conditioning on its endpoint gives the killed version of (6.2) with kernel (6.5). We justify its limit. On bounded time intervals, clock convergence implies convergence of integrals of $e^{-A_s^j} f(B_s)$ against dA_s^j for continuous f. Bridge absolute continuity gives uniform convergence of the kernels on compact sets away from the diagonal, first for durations bounded away from zero and infinity. The off-diagonal Gaussian estimate controls small durations, and exponential decay of the killed heat kernel in a ball controls large durations. Near the diagonal, $G_D(x, y) \le C|x - y|^{2-d}$, whose integral against μ_j over $B(x, r)$ is at most $Cr^{2-\delta}$, uniformly in j. The smooth killed kernels are continuous off the diagonal and vanish on the boundary; their locally uniform limit has the same properties. Weak convergence of μ_j, together with the preceding diagonal bound, gives the killed row identity for each $x \in D$. On the clock side, the error from Brownian times larger than T is at most $\|f\|_\infty \mathbb{P}_x[\tau_D > T]$, uniformly in j. Thus it vanishes as $T \to \infty$. Bounded continuous test functions determine the row measure, which proves the identity for nonnegative Borel functions too.

Increase D through concentric balls. Formula (6.5) increases to a measurable symmetric positive function r_h, and monotone convergence gives (6.2). Conservativeness holds from every starting point. To see this, couple Brownian motions from x and 0 to agree after a finite time, using successive coordinate reflection couplings. Their initial clock segments are finite by the all-start construction; additivity and the infinite total clock from 0 give an infinite total clock from x as well. Hence each row integrates to one, proving its almost-everywhere finiteness. Finally, Brownian bridge scaling and the measure covariance (1.6) give (6.3). The diagonal value is immaterial because ν_h is diffuse. □

6.2 Root exchange and the jump-Palm law

Theorem 5.8 extends to the Poisson decoration. Its random dilation is chosen from the field and the LBM path, independently of the Poisson process conditionally on the field. Conditional on the transformed field and path, the transformed process is still Poisson: its intensity is precisely the GMC of the transformed field by (1.6). Thus for every nonnegative function Ψ of a doubly rooted

decorated environment, invariant under simultaneous translations and positive dilations, LBM-time stationarity and reversal give

$$\mathbb{E}\Psi(\omega, 0, X_T^h) = \mathbb{E}\Psi(\omega, X_T^h, 0), \tag{6.6}$$

where T is an independent mean-one exponential time. Indeed, shift the two-sided process by a deterministic time t, apply its allowed dilation, and reverse time; this exchanges the two marked positions. Then integrate over t against $e^{-t}dt$.

Lemma 6.2 (Homogeneous mass transport)**.** *Suppose $\Phi(\omega, x, y) \geq 0$ is measurable and satisfies*

$$\Phi(S_{z,r}\omega, r^{-1}(x-z), r^{-1}(y-z)) = r^{d-2}\Phi(\omega, x, y),$$

where $S_{z,r}$ transforms both the field and the marked nuclei. Then, with infinite expectations permitted,

$$\mathbb{E}\int \Phi(\omega, 0, y)\,\nu_h(dy) = \mathbb{E}\int \Phi(\omega, y, 0)\,\nu_h(dy). \tag{6.7}$$

Proof. By Lemma 6.1, the conditional distribution of X_T^h has density $r_h(0, y)$ with respect to ν_h. The quotient Φ/r_h is dilation invariant. Apply (6.6) to $\min\{n, \Phi/r_h\}$, taking value zero when the denominator is not finite and positive. The exceptional set has zero mass in the row at 0. Pointwise symmetry of r_h makes the two row integrals cancel the same denominator. Monotone convergence as $n \to \infty$ proves the assertion. □

If a nonnegative vertex transport $F(\omega, u, v)$ transforms by r^{2-d} under $S_{z,r}$, apply the lemma to

$$\Phi(\omega, x, y) = \frac{F(\omega, c_\omega(x), c_\omega(y))}{m(c_\omega(x))m(c_\omega(y))}.$$

Each mass has degree $2-d$, so the quotient has the required degree $d-2$. Integration over each allocated cell cancels its mass, yielding

$$\mathbb{E}_{\mathsf{Q}_{\mathrm{cell}}}\left[\frac{1}{m(o)}\sum_v F(\omega, o, v)\right] = \mathbb{E}_{\mathsf{Q}_{\mathrm{cell}}}\left[\frac{1}{m(o)}\sum_v F(\omega, v, o)\right]. \tag{6.8}$$

Define

$$\frac{d\mathsf{Q}_{\mathrm{jump}}}{d\mathsf{Q}_{\mathrm{cell}}}(\omega, o) = \frac{1}{Z}\frac{\pi(o)}{m(o)}, \qquad Kf(\omega, o) = \sum_{v\sim o}\frac{a(o, v)}{\pi(o)}f(\theta_v\omega). \tag{6.9}$$

The quotient $\pi(o)/m(o)$ is dilation invariant by (1.6). Equation (6.1) and positivity of the cell masses and conductances make $\mathsf{Q}_{\mathrm{jump}}$ a probability measure equivalent to $\mathsf{Q}_{\mathrm{cell}}$.

Proposition 6.3 (Stationarity and reversibility)**.** *The measure $\mathsf{Q}_{\mathrm{jump}}$ is stationary and reversible for K.*

Proof. For bounded nonnegative f, insert its density to get

$$\mathbb{E}_{\mathsf{Q}_{\mathrm{jump}}}Kf = \frac{1}{Z}\mathbb{E}_{\mathsf{Q}_{\mathrm{cell}}}\left[\frac{1}{m(o)}\sum_{v\sim o}a(o, v)f(\theta_v\omega)\right].$$

Apply (6.8) with $F(\omega, u, v) = a(u, v)f(\theta_v\omega)\mathbf{1}_{\{u\sim v\}}$. Conductance symmetry gives $\mathbb{E}_{\mathsf{Q}_{\mathrm{jump}}}f$. Applying the same identity to $a(u, v)f(\theta_u\omega)g(\theta_v\omega)$ gives detailed balance. Linearity handles signed bounded functions. □

6.3 Translation-invariant functions of the decorated cone

Lemma 6.4 (Escape and shielding)**.** *For almost every decorated cone, the LBM from* 0 *satisfies*

$$\frac{1}{T}\int_0^T \mathbf{1}_D(X_t^h)\,dt \longrightarrow 0 \tag{6.10}$$

for every bounded Borel set D, including one chosen from the environment. If $R_k(z)$ is the distance from z to its kth nearest nucleus, then for every $R, a > 0$ and $k \geq 1$,

$$\{z : \operatorname{dist}(z, B_R) - R_k(z) < a\} \quad \textit{is bounded.} \tag{6.11}$$

Proof. Since $d \geq 3$, Brownian transience gives a finite Brownian time after which a bounded set is never visited. Its LBM time is finite; conservativeness then proves (6.10). First establish this simultaneously for integer-radius balls. Every bounded set, including a random bounded set after conditioning on the environment, lies in one such ball.

For each unit vector u, choose k distinct nuclei p_i with $p_i \cdot u > R + a + 2$, using the infinitely many nuclei in an open cone. These inequalities with margin 1 remain true in a neighborhood of u; finitely many neighborhoods cover the unit sphere. For $z = tv$ and a corresponding chosen nucleus p,

$$|tv - p|^2 < (t - R - a)^2 \quad \Longleftrightarrow \quad 2t(p \cdot v - R - a) > |p|^2 - (R + a)^2.$$

The finite choice makes this true for every sufficiently large t, uniformly in v, for at least k nuclei. Hence $R_k(tv) < t - R - a$, proving (6.11). □

Theorem 6.5 (Translation triviality)**.** *Let F be a bounded measurable function of continuum-rooted decorated environments modulo positive dilations. Suppose it has a version such that, on one Q_0-full event,*

$$F(\theta_z\omega) = F(\omega) \qquad \textit{for every } z \in \mathbb{R}^d. \tag{6.12}$$

Then F is Q_0-almost surely constant.

Proof. The process $E_t = \theta_{X_t^h}\omega$ is stationary on the continuum-rooted dilation quotient by Theorem 5.8, with the Poisson decoration retained as above. The continuous-time ergodic theorem, without assuming ergodicity, gives for every bounded measurable G

$$A_T G := \frac{1}{T}\int_0^T G(E_t)\,dt \longrightarrow Z_G \quad \text{almost surely and in } L^1.$$

We choose a genuinely local approximation to F. For $k \geq 2$, let $L_k(z)$ contain the field restricted to the open ball $B(z, R_k(z))$, the marked nuclei in the closed ball, and this radius, all rooted at z and normalized by the common length $R_2(z)$. These are standard Borel local data. Their sigma-fields increase and generate the rooted dilation class, since $R_k(0) \to \infty$. For any $\eta > 0$, conditional-expectation approximation therefore provides a bounded Borel function g such that

$$G(\omega) = g(L_k(0)), \qquad \mathbb{E}|F - G| < \eta, \qquad \|G\|_\infty \leq \|F\|_\infty.$$

It has the canonical jointly measurable translates $G(\theta_z\omega) = g(L_k(z))$. Stationarity and (6.12) imply

$$\mathbb{E}|F - A_T G| < \eta, \qquad \mathbb{E}|F - Z_G| \leq \eta. \tag{6.13}$$

Realize the positive LBM branch by a Brownian motion B independent of the environment and write $\mathcal{W} = \sigma(B_s : s \geq 0)$. For an integer $R \geq 1$, let $\mathcal{A}_R = \sigma(h|_{\mathbb{R}^d \setminus B_R}, \mathcal{P} \cap (\mathbb{R}^d \setminus B_R))$, retaining

the marks. Choose a continuous nondecreasing function φ equal to zero on $(-\infty, 0]$ and to one on $[1, \infty)$, and set

$$\chi_R(z) = \varphi(\operatorname{dist}(z, B_{R+1}) - R_k(z)).$$

The function R_k is 1-Lipschitz, so χ_R is continuous. Lemma 6.4 makes $1 - \chi_R$ supported in a bounded set. Both $\chi_R(z)$ and, on $\{\chi_R(z) > 0\}$, $L_k(z)$ are determined by $\mathcal{A}_R$. Indeed, putting $d_R(z) = \operatorname{dist}(z, B_{R+1})$, either fewer than k exterior nuclei lie within distance $d_R(z)$, in which case $\chi_R(z) = 0$, or the relevant first k nuclei and their entire local ball are exterior. Every nucleus in B_R is at distance at least $d_R(z) + 1$ when z lies outside B_{R+1}.

Let F_h be the Brownian clock of LBM. In Brownian time define

$$C_R^B(s) = \int_0^s \chi_R(B_u)\, dF_h(u), \qquad H_R^B(s) = \int_0^s G(\theta_{B_u}\omega)\, dC_R^B(u).$$

These processes are $\mathcal{W} \vee \mathcal{A}_R$-measurable. For the first one, its Revuz measure $\chi_R \nu_h$ is exterior measurable and vanishes on B_{R+1}; the smooth-measure construction in Lemma 6.1 constructs its PCAF measurably from that measure and B. For dC_R^B-almost every u, one has $\chi_R(B_u) > 0$, where the local data determining the second integrand are exterior measurable. Values elsewhere can be set to zero. This establishes locality of both clocks without using an unproved locality assertion about the inverse full clock.

In LBM time put

$$C_R(t) = \int_0^t \chi_R(X_u^h)\, du, \qquad D_R(t) = t - C_R(t).$$

By (6.10), $D_R(t)/t \to 0$, so C_R is unbounded. Define

$$\rho_R(T) = \inf\{s : C_R^B(s) \geq T\}, \qquad \eta_R(T) = \inf\{t : C_R(t) \geq T\}.$$

The exterior average

$$A_T^{\mathrm{ext},R} G = \frac{H_R^B(\rho_R(T))}{T} = \frac{1}{T} \int_0^{\eta_R(T)} \chi_R(X_u^h) G(E_u)\, du$$

is $\mathcal{W} \vee \mathcal{A}_R$-measurable. Continuity gives $C_R(\eta_R(T)) = T$ and $\eta_R(T) \geq T$. Moreover,

$$\int_T^{\eta_R(T)} \chi_R(X_u^h)\, du = D_R(T).$$

Consequently

$$|A_T G - A_T^{\mathrm{ext},R} G| \leq 2\|G\|_\infty D_R(T)/T \longrightarrow 0.$$

Thus Z_G is measurable modulo null sets with respect to $\mathcal{W} \vee \mathcal{A}_R$ for each integer R.

Theorem 2.5 says that the completed decreasing tail of $\mathcal{A}_R$ is trivial. Since $\mathcal{W}$ is independent of the environment,

$$\bigcap_R (\mathcal{W} \vee \mathcal{A}_R) = \mathcal{W} \quad \text{modulo null sets.}$$

For completeness, conditional expectation on this product probability space is the tensor product of the identity projection on $L^2(\mathcal{W})$ and the projection onto $L^2(\mathcal{A}_R)$; the reverse martingale theorem proves the displayed identity, including completions. Hence Z_G is $\mathcal{W}$-measurable, whereas F is independent of $\mathcal{W}$. Therefore

$$\mathbb{E}|Z_G - \mathbb{E}F| = \mathbb{E}\big|\mathbb{E}[Z_G - F \mid \mathcal{W}]\big| \leq \mathbb{E}|Z_G - F|.$$

Together with (6.13), this gives $\mathbb{E}|F - \mathbb{E}F| \leq 2\eta$. Let $\eta \downarrow 0$. □

Proposition 6.6 (Ergodicity of the jump environment)**.** *The stationary Markov chain with initial law* $\mathsf{Q}_{\mathrm{jump}}$ *and transition kernel* K *is ergodic.*

Proof. Suppose f is bounded and real-valued, with $Kf = f$ almost surely. Stationarity gives

$$\int (K(f^2) - (Kf)^2)\, d\mathsf{Q}_{\mathrm{jump}} = 0.$$

The integrand is a conditional variance. Thus from almost every rooted environment, f takes the same value after every adjacent transition of positive probability. Along a stationary chain this property holds at every integer time on a single full-probability event. Conditional on almost every initial environment, any fixed finite adjacency path has positive probability, since the edge conductances are positive and the incident sums finite. Connectivity and the countability of the vertices imply, simultaneously for every nucleus v,

$$f(\theta_v \omega) = f(\omega, o).$$

Equivalence transfers this assertion from $\mathsf{Q}_{\mathrm{jump}}$ to $\mathsf{Q}_{\mathrm{cell}}$. Under Q_0 set $F(\omega) = f(\theta_{c_\omega(0)}\omega)$. The preceding equality makes the chosen nucleus irrelevant, so $F(\theta_z \omega) = F(\omega)$ simultaneously for every $z \in \mathbb{R}^d$. Theorem 6.5 makes F, and consequently f, almost surely constant.

Finally, if A is a shift-invariant event of the stationary one-sided chain, $f(\eta_0) = \mathbb{P}(A \mid \eta_0)$ is K-harmonic. The Markov property and shift invariance give $\mathbb{P}(A \mid \eta_0, \dots, \eta_n) = f(\eta_n)$. The preceding result makes this constant; martingale convergence gives $\mathbf{1}_A = \mathbb{P}(A)$ almost surely. This is ergodicity. □

7 The clock and the quenched convergence statement

We now work in the parameter range of Theorem 1.1. The sampling measure is $M_h = \mu_{h,\gamma}$, whereas the speed measure is $\nu_h = \mu_{h,\alpha}$. These are different measures when $d > 2$. Set $\vartheta = \alpha/\gamma$ and $N_\varepsilon = \varepsilon^{-\vartheta}$. The normalizer $Z \in (0, \infty)$ is supplied by Proposition 4.4.

7.1 An ergodic clock law and an exact intensity identity

On a vertex-rooted dilation class define

$$g(\omega, o) = \frac{\nu_h(P_o)}{\pi(o)}.$$

This ratio is unchanged under the dilation covariance (1.6). Let (η_j) be the environment chain and let (E_j) be independent mean-one exponential random variables, independent of the chain.

Lemma 7.1 (Marked clock law)**.** *Under either initial law* $\mathsf{Q}_{\mathrm{jump}}$ *or* $\mathsf{Q}_{\mathrm{cell}}$*,*

$$\frac{1}{n} \sum_{j<n} g(\eta_j) E_j \longrightarrow \frac{1}{Z} \quad \textit{almost surely.}$$

The assertion also holds with E_j *replaced by* 1*.*

Proof. The jump bias cancels exactly:

$$\mathbb{E}_{\mathsf{Q}_{\mathrm{jump}}} g = \frac{1}{Z} \mathbb{E}_{\mathsf{Q}_{\mathrm{cell}}} \left[\frac{\pi(o)}{\nu_h(P_o)} \frac{\nu_h(P_o)}{\pi(o)} \right] = \frac{1}{Z}.$$

Propositions 6.3 and 6.6 make the chain stationary and ergodic. Adjoin a two-sided independent exponential sequence to a two-sided stationary extension of the chain. Its Bernoulli shift is mixing, so the product with the ergodic chain is ergodic. Birkhoff proves the marked assertion; applying it to g gives the unmarked assertion. The marked path laws from $\mathsf{Q}_{\mathrm{jump}}$ and $\mathsf{Q}_{\mathrm{cell}}$ are equivalent, since their Radon–Nikodym derivative depends only on the initial environment and is positive and finite. Both probability-one conclusions therefore transfer to $\mathsf{Q}_{\mathrm{cell}}$. □

Lemma 7.2 (Changing sampling intensity)**.** *Let $b > 0$ and sample a Poisson process $\mathcal{P}_b$ of intensity $e^{\gamma b} M_h$, conditionally on h. The field/process pair $(h + b, \mathcal{P}_b)$ rooted at the cell containing 0, and taken modulo dilation, has law $\mathsf{Q}_{\mathrm{cell}}$.*

Proof. Proposition 2.3, used with its level parameter $a = -b$, provides a field-measurable dilation of a unit-intensity decorated cone whose transformed field has law $h + b$. Poisson mapping and (1.6) give the transformed conditional intensity $\mu_{h+b,\gamma} = e^{\gamma b} M_h$. The containing cell is mapped to the containing cell, and the vertex-rooted dilation class is unchanged. Conditioning only on the transformed field preserves the Poisson assertion, since its intensity is a function of that field. □

For the discrete walk (V_j^ε) at intensity $\varepsilon^{-1} M_h$, define

$$b_\varepsilon = \gamma^{-1} \log \varepsilon^{-1}, \qquad g_j^\varepsilon = N_\varepsilon \frac{\nu_h(P^\varepsilon_{V_j^\varepsilon})}{\pi_\varepsilon(V_j^\varepsilon)}.$$

Adding b_ε multiplies the speed measure by N_ε; it does not change the transition probabilities. Lemma 7.2 and the fixed measurable kernel that samples the walk and its exponential marks give the whole-sequence identity

$$(g_j^\varepsilon, E_j)_{j \geq 0} \overset{\text{law}}{=} (g(\eta_j), E_j)_{j \geq 0}, \qquad \eta_0 \sim \mathsf{Q}_{\mathrm{cell}}. \tag{7.1}$$

This equality is under the joint environment law; it is not an identity for an individually fixed field across intensities.

Lemma 7.3 (Uniform partial sums)**.** *If $s_0 = 0$ and $s_n/n \to a$, then for every finite M,*

$$N^{-1} \max_{0 \leq n \leq MN} |s_n - an| \longrightarrow 0 \quad (N \to \infty),$$

where n is an integer and N can be real.

Proof. For any $\eta > 0$, choose n_0 so $|s_n - an| \leq \eta n$ for $n \geq n_0$. The displayed expression is at most $N^{-1} \max_{n < n_0} |s_n - an| + M\eta$. □

Couple two linear interpolations of the same discrete path by setting

$$\begin{aligned} A_n^\varepsilon &= 2 \sum_{j<n} \frac{\nu_h(P^\varepsilon_{V_j^\varepsilon})}{\pi_\varepsilon(V_j^\varepsilon)} E_j = \frac{2}{N_\varepsilon} \sum_{j<n} g_j^\varepsilon E_j, \\ C_n^\varepsilon &= \frac{2n}{Z N_\varepsilon}. \end{aligned} \tag{7.2}$$

The natural-clock path Y^ε traverses edge j on $[A_j^\varepsilon, A_{j+1}^\varepsilon]$; X^ε traverses it on $[C_j^\varepsilon, C_{j+1}^\varepsilon]$. The jump version of Y^ε has rates

$$q_\varepsilon(v, w) = \frac{a_\varepsilon(v, w)}{2\nu_h(P_v^\varepsilon)}. \tag{7.3}$$

For each fixed ε, the sequence identity and marked law imply $A_n^\varepsilon \to \infty$ almost surely, which establishes nonexplosion.

Proposition 7.4 (Functional clock comparison)**.** *For every finite M,*

$$\max_{n\le MN_\varepsilon} |A_n^\varepsilon - C_n^\varepsilon| \longrightarrow 0$$

in joint probability. There are increasing piecewise linear homeomorphisms λ_ε of $[0,\infty)$ such that $X^\varepsilon = Y^\varepsilon \circ \lambda_\varepsilon$ and

$$\sup_{t\le T} |\lambda_\varepsilon(t) - t| \longrightarrow 0 \quad \textit{in probability for every } T < \infty.$$

Proof. Apply Lemma 7.3 to $s_n = \sum_{j<n} g(\eta_j)E_j$ and $a = 1/Z$, and transfer the resulting almost-sure estimate at $N = N_\varepsilon$ by (7.1). This yields the first assertion in probability. Set $\lambda_\varepsilon(C_n^\varepsilon) = A_n^\varepsilon$ and interpolate linearly. Positivity of the increments and divergence of both clocks give a homeomorphism. For fixed T, choose $M > ZT/2 + 1$. Then the deterministic endpoints C_n^ε, $n \le MN_\varepsilon$, cover $[0,T]$ for small ε. The maximal difference from the identity occurs at endpoints of the linear pieces, proving the second assertion. □

7.2 A local finite-graph elliptic estimate

We give the finite-graph argument underlying [3, Theorem 5.4], using the diameter of Q_v throughout. This makes explicit the mesh needed when the argument is applied recursively to subsets of vertices. In particular, it does not require the entire cells of neighboring vertices to have the smaller mesh of the current subset.

For closed geometric cells, set

$$Q_{v,w} = \operatorname{conv}(v, \overline{P}_v \cap \overline{P}_w) \cup \operatorname{conv}(w, \overline{P}_v \cap \overline{P}_w), \qquad Q_v = \overline{P}_v \cup \bigcup_{w\sim v} Q_{v,w}.$$

For a finite vertex set A, let $E[A]$ contain all edges with at least one endpoint in A, let ∂A be its exterior vertex boundary, and put $P_A = \bigcup_{v\in A} \overline{P}_v$ and $Q_A = \bigcup_{v\in A} Q_v$. All cells of A are complete geometric cells, and all their facets are paired with the indicated neighbors. Their allocated boundaries have zero mass for the measures below. Write

$$\mathcal{E}_A(f) = \sum_{e\in E[A]} a(e)|\nabla f(e)|^2, \qquad h_A = \max_{v\in A} \operatorname{diam}(Q_v).$$

Empty sets are omitted from maxima and terminate the constructions below. All cells, facets, and line segments joining adjacent nuclei under consideration lie in a fixed convex open set on which the continuous functions are defined, and the underlying tiling covers that set.

Lemma 7.5 (Local energy estimate)**.** *Let m be a nonnegative Radon measure and ζ a signed Radon measure, with m and $|\zeta|$ giving cell boundaries zero mass. Suppose $u \in C^{1,\beta}$, $0 < \beta \le 1$, satisfies $-\Delta u + um = \zeta$ in distributions on a neighborhood of the cells. Put $H = [\nabla u]_{C^{0,\beta}}$ and $L = \|\nabla u\|_\infty$ there. If $u_A = u$ on ∂A and*

$$-\Delta_a u_A(v) + m(P_v)u_A(v) = \zeta(P_v), \qquad v \in A,$$

then, for $e_A = u_A - u$ evaluated at nuclei,

$$\mathcal{E}_A(e_A) + \sum_{v\in A} m(P_v)e_A(v)^2 \le dH^2 h_A^{2\beta} \operatorname{Vol}_d(Q_A) + L^2 h_A^2 m(P_A). \tag{7.4}$$

Proof. For an oriented edge from v to w, let F_{vw} be its facet, $\ell_{vw} = |w-v|$, and $n_{vw} = (w-v)/\ell_{vw}$. The exact continuous flux is $\int_{F_{vw}} \nabla u \cdot n_{vw}\, d\sigma$. Its difference r_{vw} from $a(v,w)(u(w)-u(v))$ satisfies

$$|r_{vw}| \le H h_A^{\beta} \operatorname{Vol}_{d-1}(F_{vw}) \quad \text{if } v \in A \text{ or } w \in A.$$

Indeed, if $v \in A$, then $v, w, y \in Q_v$ for every $y \in F_{vw}$, so for $0 \le t \le 1$, $|y-((1-t)v+tw)| \le \operatorname{diam}(Q_v)$. Integrating the gradient along $[v,w]$ proves the claim. The argument does not assert that $[v,w]$ lies in Q_v.

The divergence identity on each cell follows by mollifying u: its gradients converge uniformly, and the measures of cells converge because the boundaries have zero mass. Subtract this identity from the discrete equation and multiply by $e_A(v)$. Discrete summation by parts, followed by $ab \le (a^2+b^2)/2$, bounds the resulting energy by one half of itself plus

$$\frac{1}{2} \sum_{\{v,w\} \in E[A]} H^2 h_A^{2\beta} \ell_{vw} \operatorname{Vol}_{d-1}(F_{vw}) + \frac{1}{2} L^2 h_A^2 m(P_A).$$

Here the mass error uses $|\int_{P_v} (u(y)-u(v))\, dm(y)| \le L h_A m(P_v)$. Finally, $\ell_{vw} \operatorname{Vol}_{d-1}(F_{vw}) = d \operatorname{Vol}_d(Q_{v,w})$. The double pyramids for distinct edges have disjoint interiors and their union is contained in Q_A. Rearrangement gives (7.4). □

Proposition 7.6 (Finite elliptic convergence under the shape condition)**.** *In the preceding setting, suppose that the sets Q_A lie in a fixed bounded region, that $m(P_A) \le M$, and that*

$$\operatorname{diam}(Q_v) \le J \operatorname{Vol}_d(Q_v)^{\rho}, \qquad v \in A, \qquad J, \rho > 0.$$

There are $C < \infty$ and $h_0 > 0$, depending only on that region, $d, \beta, H, L, M, J, \rho$, such that, if $0 < h_A \le h_0$,

$$\max_{v \in A} |u_A(v) - u(v)| \le C h_A^{\beta} (1 + \log(h_A^{-1})). \tag{7.5}$$

Proof. Two elementary estimates for finite orthogonal tilings are used; compare [4, Lemmas 2.4, 3.1, and 3.5]. If $f = 0$ on ∂S, then

$$\|f\|^2_{L^2(P_S)} \le D^2 \mathcal{E}_S(f), \qquad D = \operatorname{diam}(Q_A), \quad S \subset A.$$

To see this, slice the tiling by lines in a coordinate direction. The projections of consecutive nuclei along a line increase, by orthogonality. Telescoping f from its zero exterior value and Cauchy–Schwarz give a bound by the total projected edge length times the sum of squared increments divided by those lengths. The total projected length is at most D; gaps between separate components of S only decrease it. Integrating over the transverse coordinates converts the projected facet areas to the conductances. A further integration along each line contributes at most D. This also proves the estimate when S is disconnected. Second, if $h \ge \max_{v \in S} \operatorname{diam}(Q_v)$, then

$$d \sum_{e \in E[S]\colon |\nabla f(e)| > h} \operatorname{Vol}_d(Q_e) \le \mathcal{E}_S(f),$$

since $|e| \le h$ and $a(e) h^2 \ge a(e)|e|^2 = d \operatorname{Vol}_d(Q_e)$.

Set $S_{-1} = A$, $u_0 = u_A$, and $h_0' = h_A$. Given u_i and h_i', define

$$N_i = \{v \in A : |u_i(v) - u(v)| > T (h_i')^{\beta}\},$$

and let S_i be obtained from N_i by removing each vertex having a neighbor outside N_i across which $|\nabla(u_i - u)| \le h_i'$. Let u_{i+1} solve the same inhomogeneous discrete equation on S_i, with boundary

values u and with $u_{i+1} = u$ outside S_i. Set $h'_{i+1} = \max_{v\in S_i} \operatorname{diam}(Q_v)$. Then $S_{i+1} \subset N_{i+1} \subset S_i$, and the maximum principle gives

$$\|u_{i+1} - u_i\|_\infty \le h'_i + T(h'_i)^\beta.$$

The discrete Dirichlet solutions exist uniquely, for example by the preceding Poincare inequality and positivity of the quadratic form.

Put $A_i = \max\{\operatorname{Vol}_d(Q_{S_i}), m(P_{S_i})\}$. For $f = u_{i+1} - u$, Lemma 7.5, applied on S_i with precisely the mesh h'_{i+1}, yields, when $h_A \le 1$,

$$\mathcal{E}_{S_i}(f) + \sum_{v\in S_i} m(P_v) f(v)^2 \le c(h'_{i+1})^{2\beta} A_i, \qquad c = dH^2 + L^2.$$

Markov's inequality and the Poincare estimate bound $\operatorname{Vol}_d(P_{N_{i+1}})$ by cD^2A_i/T^2 and $m(P_{N_{i+1}})$ by cA_i/T^2. The part of $Q_{S_{i+1}}$ outside $P_{N_{i+1}}$ lies in double pyramids of edges from S_{i+1} to the complement of N_{i+1}. Their f-increments exceed h'_{i+1} by construction, so their total volume is at most $c(h'_{i+1})^{2\beta}A_i/d$. Consequently

$$A_{i+1} \le \left(\frac{c(D^2+1)}{T^2} + \frac{ch_A^{2\beta}}{d} \right) A_i.$$

Choose T large and then h_0 small so the coefficient is at most $1/2$; the case $c = 0$ follows immediately from the energy estimate. Thus $A_i \le M_0 2^{-i}$, where M_0 bounds both the volume of the fixed region and M. The shape assumption now gives

$$h'_{i+1} \le h_A \wedge JM_0^\rho 2^{-i\rho}.$$

Every vertex eventually leaves the decreasing bad sets, since each cell has positive volume. Telescoping the maximum-principle bounds therefore gives

$$\|u_A - u\|_\infty \le (T+1) \sum_{i\ge 0} (h'_i)^\beta \le C h_A^\beta (1 + \log(h_A^{-1})),$$

where the last bound follows by splitting the geometric series where $JM_0^\rho 2^{-i\rho}$ crosses h_A. This proves the claim. □

7.3 The normalized finite-tiling exit kernel

We apply Proposition 7.6 to the exact discrete Feynman–Kac equation. This fixes the Brownian time convention directly.

Lemma 7.7 (Normalized joint exit kernel)**.** *Let $B = B(a, R)$ have closure in an open set O. Let μ be a Radon measure on O, with full support and*

$$\mu(B(x, r)) \le Cr^{d-\delta}, \qquad 0 \le \delta < 1,$$

on a fixed neighborhood of $\overline{B}$. Suppose its Brownian PCAF A is strict and strictly increasing before leaving that neighborhood. Let finite orthogonal tilings cover this neighborhood, with complete cells and their required neighbors contained in O, positive finite cell masses, and local mesh $\epsilon_n \to 0$. Assume

$$\operatorname{diam}(Q_v) \le J \operatorname{Vol}_d(Q_v)^\rho \tag{7.6}$$

for fixed $J, \rho > 0$ and all relevant cells, and $\mu(\partial \overline{P}_v) = 0$.

Let Z^n be the step chain with rates $a(v,w)/(2\mu(P_v))$. Put

$$V_n(B) = \{v : \overline{P}_v \cap B \neq \varnothing\}, \qquad T_n = \inf\{t : Z^n_t \notin V_n(B)\}.$$

For standard Brownian motion W, put $\tau_B = \inf\{s : W_s \notin B\}$. If $|v_n(x) - x| = O(\epsilon_n)$, then for every compact $K \subset B$,

$$\sup_{x \in K} d_{\Pr}\Big(\mathrm{Law}_{v_n(x)}(Z^n_{T_n}, T_n), \mathrm{Law}_x(W_{\tau_B}, A_{\tau_B})\Big) \longrightarrow 0.$$

The conclusion holds simultaneously for any finite family of such balls. The exit vertex lies within $O(\epsilon_n)$ of ∂B, and the chain before exit stays within $O(\epsilon_n)$ of $\overline{B}$.

Proof. The mesh and neighbor assumptions give the last two assertions. All stopping times here are vertex stopping times of a finite graph, so no rounding is needed to apply the strong Markov property. We may increase δ slightly within $(0,1)$ when using local potential regularity.

Fix $\lambda \geq 0$ and $g \in C^\infty_c(\mathbb{R}^d)$. Conditioning on the first exponential holding time shows that

$$u_n(v) = \mathbb{E}_v[g(Z^n_{T_n})e^{-\lambda T_n}]$$

solves, with $u_n = g$ on the graph boundary of $V_n(B)$,

$$-\Delta_a u_n(v) + 2\lambda\mu(P_v)u_n(v) = 0. \tag{7.7}$$

The correctly normalized Brownian Feynman–Kac formula says that

$$u(x) = \mathbb{E}_x[g(W_{\tau_B})e^{-\lambda A_{\tau_B}}]$$

solves $-\Delta u + 2\lambda u\mu = 0$ in B, with boundary data g. The factor two corresponds to Brownian generator $\frac{1}{2}\Delta$.

For fixed small $\eta > 0$, let $B_\eta = B(a, R+\eta)$ and define u^η using exit from B_η in place of B. The decay condition gives local $C^{1,1-\delta}$ regularity of u^η. Solve (7.7) on the *same* set $V_n(B)$ with boundary data u^η, and call the solution w^η_n. Apply Proposition 7.6 with potential $2\lambda\mu$, forcing zero, and ambient domain compactly contained in B_η and containing $\overline{B}$. For each fixed η, all cells and relevant edge segments lie in the interior regularity region for large n, and the proposition gives

$$\sup_{v \in V_n(B)} |w^\eta_n(v) - u^\eta(v)| \longrightarrow 0.$$

The discrete maximum principle bounds $|u_n - w^\eta_n|$ by $\sup_{v \in \partial V_n(B)} |g(v) - u^\eta(v)|$. The graph boundary lies in the $O(\epsilon_n)$ outer annulus of B, so its limiting bound is $\sup_{x \in \partial B} |g(x) - u^\eta(x)|$.

This bound tends to zero with η. Indeed, for $x \in \partial B$,

$$|u^\eta(x) - g(x)| \leq \|\nabla g\|_\infty \mathbb{E}_x |W_{\tau_{B_\eta}} - x| + \lambda \|g\|_\infty \mathbb{E}_x A_{\tau_{B_\eta}}.$$

The first expectation is $O(\sqrt{\eta})$. The killed Brownian Green bound

$$G_{B_\eta}(x,y) \leq C \operatorname{dist}(x, \partial B_\eta)|x-y|^{1-d}$$

and annular summation using $\delta < 1$ bound the second expectation by $C\eta$. The constants are uniform for small η. The continuous maximum principle, or strong Markov at τ_B, also gives $\sup_B |u^\eta - u| \to 0$. First taking $n \to \infty$ for fixed η, and then $\eta \downarrow 0$, proves

$$\sup_{x \in K} \Big| \mathbb{E}_{v_n(x)} g(Z^n_{T_n}) e^{-\lambda T_n} - \mathbb{E}_x g(W_{\tau_B}) e^{-\lambda A_{\tau_B}} \Big| \longrightarrow 0.$$

Continuity of u handles the $O(\epsilon_n)$ start approximation.

For time tightness take a fixed larger concentric ball B_* whose closure lies in the neighborhood where the mass bound holds, and let $w(x) = \mathbb{E}_x A_{\tau_{B_*}}$. It is bounded and locally $C^{1,1-\delta}$ near $\overline{B}$, is nonnegative, and solves $-\Delta w = 2\mu$. Let w_n solve $-\Delta_a w_n(v) = 2\mu(P_v)$ on $V_n(B)$, with boundary data w. Proposition 7.6, now with zero potential and forcing 2μ, gives $w_n \to w$ uniformly there. The mean exit time $t_n(v) = \mathbb{E}_v T_n$ satisfies the same equation with zero boundary values; hence $0 \le t_n \le w_n$. This proves uniform time tightness. Exit positions belong to a fixed compact set.

The products $g(y)e^{-\lambda t}$ determine probability laws on that compact spatial set times $[0,\infty)$. Their convergence and the preceding tightness therefore give joint weak convergence. It is uniform over K: otherwise take offending $x_n \in K$, extract $x_n \to x$, and use the transform identity to identify every tight subsequential joint limit with the kernel at x. The target kernel is continuous in its start by the same Feynman–Kac regularity, time bound, and transform characterization. The resulting contradiction proves uniformity. A finite family of balls is handled by taking the maximum of its finitely many errors. □

7.4 From local exit kernels to global paths

The following coupling uses exact vertex stopping times and compact containment on each finite time interval.

Proposition 7.8 (Local exit kernels determine the path limit)**.** *Let X be a continuous conservative time change of standard Brownian motion on $\mathbb{R}^d$, started at x, by a strict, continuous, strictly increasing PCAF. Assume that this clock is finite at the Brownian exit time of every bounded ball. Let Z^n be nonexplosive continuous-time vertex chains, started at $x_n \to x$, and let Y^n be their linear interpolations between successive jump times.*

Suppose that on every compact set the maximal jump length tends to zero and that the following local assertions hold. For every ball $B = B(a,R)$ there is a vertex stopping time T_B^n such that the interpolated segment up to T_B^n stays in $B(a, R + o_n(1))$, and its terminal vertex is within $o_n(1)$ of ∂B. The errors are uniform over starts in a compact subball. The joint kernels of terminal vertex and duration converge uniformly over those starts to the corresponding exit kernel of X. The latter kernel is weakly continuous in its starting point. These assertions are required simultaneously only for each fixed finite family of balls.

Then Y^n converges in law to X in $C([0,\infty),\mathbb{R}^d)$ with the local uniform topology. The step chains have the same limit in local Skorokhod topology. In particular, Lemma 7.7 supplies the local hypotheses for the rates $a(v,w)/(2\mu(P_v))$, whenever its assumptions hold locally on every compact set.

Proof. Fix a time horizon $T < \infty$ and an error probability $\eta > 0$. Continuity and conservativity allow a compact ball K, with x in its interior, such that

$$\mathbb{P}_x\{X[0,T+1] \not\subset \operatorname{int} K\} < \eta. \tag{7.8}$$

Choose $r > 0$ small and a finite $r/2$-net $\{a_i\}$ of K, with centers in K. All balls $B_i = B(a_i, 2r)$ lie in a fixed bounded neighborhood of K. Starting from $y_0 = x$ and $t_0 = 0$, choose at stage k the first center satisfying $|y_k - a_i| \le r/2$, and let t_{k+1} be the exit time from that ball after t_k; put $y_{k+1} = X_{t_{k+1}}$. Stop at the first index N with $y_N \notin K$.

This index is finite almost surely. Indeed, use the corresponding Brownian exit times β_k. Each completed increment has conditional mean at least $(4 - 1/4)r^2/d$, by the Brownian mean exit-time formula. All segments before the last anchor stay in the $3r$-neighborhood of K. Consequently $\beta_{N\wedge m}$

is bounded by Brownian exit from a fixed larger ball, and summing the conditional mean bound gives $\mathbb{E}(N \wedge m) \le C_K/r^2$. Letting $m \to \infty$ proves $N < \infty$. The assumed local clock finiteness also gives $t_N < \infty$. Choose a deterministic J so that $\mathbb{P}(N > J) < \eta$.

We couple at most J successive segments. The ball at every stage is chosen from the *continuum* anchor, and the same ball is used for both processes. Thus no continuity of a grid-selection rule is needed. On successful previous couplings the discrete anchor is within $o(1)$ of the continuum anchor, so both belong to the compact central subball $\overline{B}(a_i, r)$.

Here are details of the sequential errors. Write $e_n \to 0$ for the largest local kernel error over this finite family, and let $\omega_r(s) \to 0$ as $s \downarrow 0$ be a common modulus of weak continuity of the limiting kernels on their compact central subballs. Use Prokhorov distance for the bounded product metric $1 \wedge (|z - z'| + |u - u'|)$ on exit position and duration. Starting with $b_{n,0} = |x_n - x|$, set

$$b_{n,k} = e_n + \omega_r(b_{n,k-1}) + n^{-1}, \qquad 1 \le k \le J.$$

For fixed J, all these numbers tend to zero. The coupling characterization of Prokhorov distance, applied measurably to the successive kernels, couples the kth positions and durations with error at most $b_{n,k}$, except on an event of conditional probability at most $b_{n,k}$. Use arbitrary couplings after the first failure. Regular conditional laws of the complete stopped segments given their exit data, and the strong Markov property, realize this as a coupling of the original paths. In particular, sampling only exit data does not replace either process by a different interpolation. The union of the coupling failures has probability $o(1)$.

Put $b_n = \max_{k \le J} b_{n,k}$. Write s_k for the discrete anchor times. On the event $N \le J$ and no coupling failure,

$$\max_{k \le N} |s_k - t_k| \le J b_n.$$

Every duration is strictly positive. Therefore the map $\lambda_n(t_k) = s_k$, interpolated linearly between anchors and extended by slope one after t_N, is an increasing homeomorphism, and

$$\sup_{t \ge 0} |\lambda_n(t) - t| \le J b_n. \tag{7.9}$$

The difference from the identity is linear on each interval, so only its endpoint values enter this estimate. No smallest-duration estimate is used.

Freeze auxiliary copies $\overline{X}$ and $\overline{Y}^n$ at their last paired anchors t_N and s_N. Within paired segments both paths stay in the same ball of radius $2r + o(1)$. After the last segment their anchors are within b_n. Hence

$$\sup_{t \ge 0} |\overline{Y}^n(\lambda_n(t)) - \overline{X}(t)| \le 4r + o(1) + b_n.$$

Together with (7.9), this bounds the distance at equal times on $[0, T]$ by

$$4r + o(1) + b_n + \omega_X(T + 1, J b_n),$$

for large n, where $\omega_X(L, h) = \sup\{|X_u - X_v| : u, v \le L,\ |u - v| \le h\}$. Stopping a continuous path cannot increase this modulus.

On the event in (7.8)'s complement, $t_N > T + 1$; for large n also $s_N > T$. Thus neither auxiliary freezing changes either path on $[0, T]$. Since $\omega_X(T + 1, J b_n) \to 0$ in probability (the continuum marginal is fixed), we have constructed couplings such that

$$\limsup_{n \to \infty} \mathbb{P}\left\{ \sup_{t \le T} |Y_t^n - X_t| > 5r \right\} \le 2\eta.$$

First let $r \downarrow 0$ and then $\eta \downarrow 0$. This gives weak convergence on every finite time interval, and hence in the local uniform topology. On the same compact-containment event the distance between the jump path and its interpolation is bounded by the maximal local jump length. This proves the final assertion. □

7.5 Poisson mesh, shape, and the natural clock

Lemma 7.9 (Local geometry uniformly across intensities)**.** *Fix a field in the full-probability mass-bound event of Lemma 3.1. Couple intensity-mM_h processes, $m \geq 1$, by a Poisson process on $\mathbb{R}^d \times [0,\infty)$ of intensity $M_h(dx)\,dt$. Almost surely, on every fixed compact set, the cells and their neighbors have mesh tending to zero and satisfy (7.6), with constants independent of all sufficiently large real m.*

Proof. All probabilities in this proof are conditional on the field. On a fixed compact ball, enlarged whenever needed below, the mass upper bound and the two-point Poisson formula give

$$\mathbb{P}(\exists v \neq w \in \mathcal{P}_m \cap K : |v - w| < r \mid h) \leq C_h m^2 r^{d-\delta}.$$

Choose J with $J(d-\delta) > 3$, apply this at $m = 2^{n+1}$ and $r = 2^{-Jn}$, and use Borel–Cantelli and monotonicity. Eventually all relevant distinct nuclei are separated by at least cm^{-J}, simultaneously for real m.

The uniform lower mass bound $M_h(B(z,r)) \geq c_h r^\ell$ from Lemma 3.1 and a finite grid give

$$\mathbb{P}\left(\sup_{x \in K} \operatorname{dist}(x, \mathcal{P}_m) > Cr \mid h\right) \leq Cr^{-d} \exp(-c_h m r^\ell).$$

Choose $\zeta > 0$ with $\zeta\ell < 1$, apply this at $m = 2^n$ and $r = 2^{-\zeta n}$, and again use Borel–Cantelli and monotonicity. The covering radius is eventually at most $C'_h m^{-\zeta}$ for all real m.

Apply the latter estimate on nested compact balls. A cell meeting the innermost ball has its nucleus within the covering radius; convexity then confines the whole cell to a ball of comparable radius about its nucleus. Otherwise the line segment to a distant point of the cell would include a point in the larger ball farther than the covering radius from its nearest nucleus. Neighboring nuclei are within twice this radius. The same argument on the next larger ball gives

$$\operatorname{diam}(Q_v) \leq C''_h m^{-\zeta}.$$

The separation estimate supplies an inscribed ball of radius cm^{-J} in each relevant cell, so $\operatorname{Vol}_d(Q_v) \geq c' m^{-Jd}$. These inequalities prove (7.6) with $\rho = \zeta/(Jd)$. A countable compact exhaustion gives all the simultaneous local assertions. □

Let $\mathcal{C} = C([0,\infty), \mathbb{R}^d)$ have the bounded local uniform metric

$$d_{\mathrm{lu}}(x,y) = \sum_{k \geq 1} 2^{-k} \left(1 \wedge \sup_{t \leq k} |x(t) - y(t)|\right),$$

and let d_{BL} be its bounded-Lipschitz distance on probability laws, using tests $|f| \leq 1$ and $\operatorname{Lip}(f) \leq 1$.

Proposition 7.10 (Natural-clock conditional convergence)**.** *With Y^ε defined by (7.2),*

$$d_{\mathrm{BL}}\left(\operatorname{Law}(Y^\varepsilon \mid h, \mathcal{P}_\varepsilon), \operatorname{Law}(X \mid h)\right) \longrightarrow 0$$

in probability under the joint field/Poisson law.

Proof. Consider an arbitrary deterministic sequence $\varepsilon_n \downarrow 0$. Fix a field with the mass and strict full-support clock properties, and with almost-sure natural-walk nonexplosion at every intensity in this sequence. Disintegration and the nonexplosion established after (7.3) give a full-probability set of such fields. Use the preceding Poisson-rain coupling.

For every fixed ball B, retain all complete cells meeting a larger compact neighborhood and one layer of their neighbors. Local mesh convergence places all these cells in a still larger fixed compact neighborhood. This gives a finite orthogonal tiling with exactly the full tessellation's outgoing edges and speed masses for the exit problem from B. No cell is clipped at a curved boundary, and ν_h is defined on all the retained cells. Lemma 7.9 verifies mesh and shape, and Lemma 3.1 verifies the local measure hypothesis. Lemma 7.7 therefore gives the uniform joint exit kernels on each fixed ball and simultaneously on finite families. The nearest nucleus to 0 is its containing cell's nucleus, outside the null tie event, and tends to 0 by local mesh convergence.

The all-start construction in Lemma 6.1 identifies the strict clocks, and Lemma 5.1 gives a continuous conservative limit. Natural-walk nonexplosion was fixed at the beginning of the proof. Proposition 7.8 now gives conditional local uniform convergence under this Poisson-rain coupling, without a separate boundary-stopping theorem.

The kernel distance is a function of the single-intensity pair $(h, \mathcal{P}_\varepsilon)$, so its conditional probability of exceeding a fixed threshold is independent of the coupling across intensities. Integrating over the field and using bounded convergence gives joint convergence in probability along the arbitrary sequence. The sequential criterion gives the full assertion. □

7.6 Conditional laws and the deterministic step clock

Lemma 7.11 (Conditional coupling and near-identity clocks)**.** *If coupled $\mathcal{C}$-valued variables U_n, W_n satisfy $d_{\mathrm{lu}}(U_n, W_n) \to 0$ in probability, then for any sigma-fields $\mathcal{F}_n$,*

$$d_{\mathrm{BL}}(\mathrm{Law}(U_n \mid \mathcal{F}_n), \mathrm{Law}(W_n \mid \mathcal{F}_n)) \to 0$$

in probability. Moreover, if the laws of W_n are tight and $\sup_{t \le T} |\lambda_n(t) - t| \to 0$ in probability for every finite T, then $d_{\mathrm{lu}}(W_n \circ \lambda_n, W_n) \to 0$ in probability.

Proof. The first distance is bounded by $\mathbb{E}[d_{\mathrm{lu}}(U_n, W_n) \mid \mathcal{F}_n]$. Boundedness turns convergence in probability into convergence in L^1, and Markov's inequality proves the claim. For the second, restrict W_n with arbitrarily high probability to a compact set in $C([0, T+1], \mathbb{R}^d)$; its common modulus of continuity tends to zero by Arzelà–Ascoli. On $\sup_{t \le T} |\lambda_n(t) - t| \le r < 1$, that modulus at r bounds the uniform error. Let $n \to \infty$, then $r \downarrow 0$, and exhaust T. □

Completion of Theorem 1.1. Proposition 4.4 gives $0 < Z < \infty$. Write

$$\widetilde{K}_\varepsilon = \mathrm{Law}(Y^\varepsilon \mid h, \mathcal{P}_\varepsilon), \quad K_\varepsilon = \mathrm{Law}(X^\varepsilon \mid h, \mathcal{P}_\varepsilon), \quad K_h = \mathrm{Law}(X \mid h).$$

Proposition 7.10 gives $d_{\mathrm{BL}}(\widetilde{K}_\varepsilon, K_h) \to 0$ in joint probability and hence in L^1. The mixture inequality for bounded-Lipschitz distance gives annealed weak convergence of Y^ε to X, and in particular tightness. Proposition 7.4 and Lemma 7.11 give

$$d_{\mathrm{lu}}(X^\varepsilon, Y^\varepsilon) \longrightarrow 0 \quad \text{in joint probability.}$$

Apply the same lemma with $\mathcal{F}_\varepsilon = \sigma(h, \mathcal{P}_\varepsilon)$ and use the triangle inequality. It follows that

$$D_\varepsilon := d_{\mathrm{BL}}(K_\varepsilon, K_h) \longrightarrow 0 \quad \text{in joint probability.}$$

For the field-only kernel, the mixture inequality gives

$$d_{\mathrm{BL}}(\mathrm{Law}(X^{\varepsilon} \mid h), K_h) \le \mathbb{E}[D_{\varepsilon} \mid h] \longrightarrow 0 \quad \text{in probability over } h.$$

Boundedness and Markov's inequality justify the last implication. These assertions imply the corresponding Prokhorov statements. Quantitatively, for the bounded metric above, $d_{\mathrm{Pr}} \le \sqrt{d_{\mathrm{BL}}}$ when $d_{\mathrm{BL}} \le 1$: test a closed set with $a(1 - a^{-1}\,\mathrm{dist}(\cdot, F))_+$ and $a = \sqrt{d_{\mathrm{BL}}}$.

A step path differs from its interpolation by at most its largest traversed edge. Tightness localizes the path to a compact ball with arbitrarily high probability, and local mesh convergence makes this edge length vanish there. Conditional coupling transfers both kernel conclusions to step paths in the local Skorokhod J_1 topology. □

Remark 7.12 (The exact quenched quantifier). The conclusion is convergence in probability of conditional laws. The sequence identity (7.1) supplies no summable cross-intensity rate and does not prove convergence for almost every fixed field along all intensities. No such stronger conclusion is claimed here.

Provenance and consolidation

The starting point was the revised September 8, 2026 manuscript *Quenched convergence in probability of Poisson–Voronoi random walk on a quantum cone*, rather than its older 24-page draft. The revision's level averaging, finite-volume cancellation, all-start resolvent, and ergodic clock argument are retained. We use the standard Brownian generator $\frac{1}{2}\Delta$, prove the corresponding finite-volume exit equation with coefficient $2\lambda\mu$, and give a direct exit-kernel-to-path argument instead of importing the older parameterized-walk draft's path coupling. The present manuscript integrates those arguments with the odd-dimensional cone construction and the localized Stieltjes-Palm exterior-tail argument. In particular, a conditional Markov restart at a negative cone passage is neither stated nor used. A separate version inventory records the alternative PDFs and duplicates reviewed during consolidation.

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